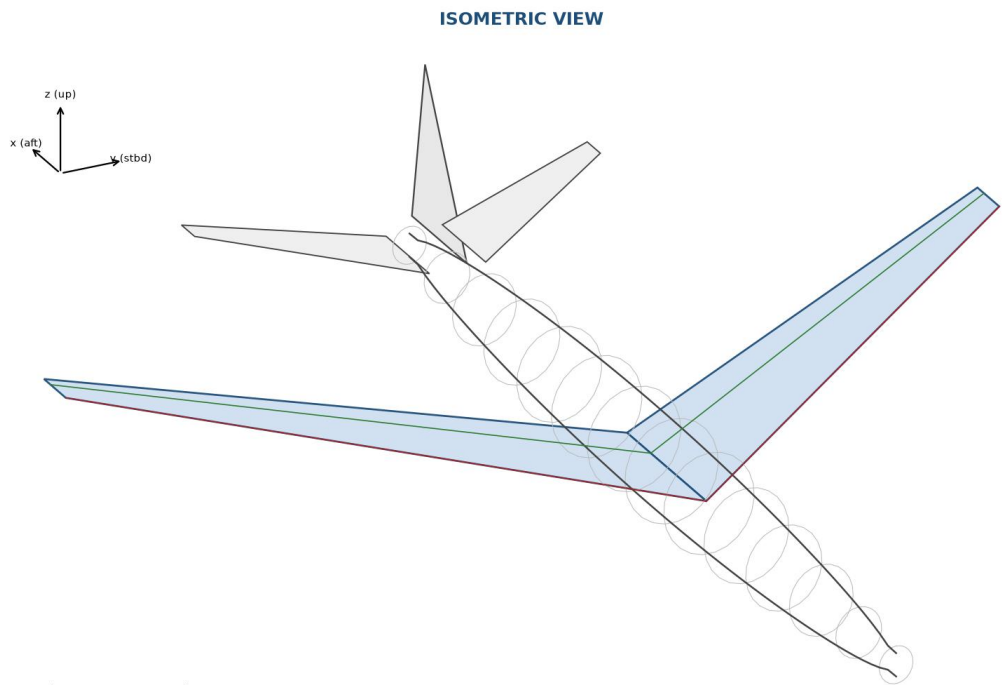


COMPUTATIONAL AERODYNAMIC CASE STUDY

RANS CFD Analysis of the NASA High-Lift Common Research Model (CRM-HL) Wing-Body across an Angle-of-Attack Sweep using ANSYS Fluent driven by Python (PyFluent)



AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
ISOMETRIC	DWG A-105
Units: m (airfoil: mm)	SCALE NTS

A self-contained engineering case study of subsonic high-lift aerodynamics

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Companion drawing set: aero.dwg.docx (DWG A-100 ... A-105)

Simulation platform: ANSYS Fluent 2025 R2 via ansys-fluent-core (PyFluent), local workstation

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1 Executive Summary

This case study is a single, coherent, end-to-end engineering analysis of the subsonic high-lift aerodynamics of the NASA High-Lift Common Research Model in wing-body configuration (CRM-HL-WB) — a publicly defined transport research geometry in its landing configuration with the leading-edge slat and trailing-edge flap deployed. The analysis follows established high-lift CFD best practice end-to-end: geometry, mesh, solver set-up, the angle-of-attack sweep, and the full aerodynamic post-processing.

The physical problem is the Reynolds-Averaged Navier-Stokes (RANS) prediction of the aerodynamic force system on the lifting body as a function of angle of attack (AoA). The simulation is implemented in Python through PyFluent, executed on a local ANSYS Fluent installation. The primary deliverable is a converged AoA sweep (0° – 14.5°) yielding lift, drag and pitching-moment coefficients, the drag polar, and the lift-to-drag ratio, with a critical (stall) angle of attack of $\alpha \approx 13.5^\circ$ and a maximum lift coefficient $C_{L,max} \approx 1.193$, followed by the full standard aerodynamic-analysis suite (Section 13).

Table 1.1 — Headline results of the primary CRM-HL-WB case.

Key performance indicator	Value	Note
Maximum lift coefficient, $C_{L,max}$	1.193	at $\alpha = 13.5^\circ$ (critical/stall angle)
Minimum drag coefficient, $C_{D,min}$	0.01576	at $\alpha = 0^\circ$
Maximum lift-to-drag ratio, $(L/D)_{max}$	≈ 20.7	at $\alpha \approx 4^\circ$
Pitching-moment behaviour	$-0.060 \rightarrow +0.143$	nose-down to nose-up reversal past stall
Freestream Mach number	0.20	incompressible-to-locally-compressible
Chord Reynolds number	5.6×10^6	based on MAC = 7.005 m
Selected turbulence model	Spalart-Allmaras	k- ω SST verified as alternative

2 Introduction, Background and Problem Statement

High-lift aerodynamics — the behaviour of a wing fitted with deployed leading-edge slats and trailing-edge flaps in the take-off and landing configuration — sets the low-speed performance boundaries of a transport aircraft: the maximum lift coefficient, the stall angle, the approach speed and the climb gradient. Predicting it with computational fluid dynamics is demanding because the flow is dominated by strong adverse pressure gradients, confluent boundary layers and wakes, slat/flap-gap flows, wing-tip vortices and the onset of separation near stall.

This study analyses one such configuration — the NASA High-Lift Common Research Model in its wing-body form (CRM-HL-WB), a publicly defined research geometry — using a steady Reynolds-Averaged Navier-Stokes (RANS) method in ANSYS Fluent, driven from Python (PyFluent). The aircraft is swept through angle of attack from 0° to beyond stall, and the resulting lift, drag and pitching-moment coefficients, the drag polar, and the full set of derived aerodynamic and performance parameters are computed and analysed.

The methodology adopted here follows established best practice for high-lift CFD: a pressure-based RANS solver, a wall-resolved mesh ($y^+ \approx 1$) with prism boundary layers, a one-equation Spalart-Allmaras eddy-viscosity turbulence model (with the $k-\omega$ SST model as a verified alternative), the energy equation retained for local compressibility, a formal grid-independence study, and an angle-of-attack sweep mapping the force and moment curves up to and beyond the stall.

2.1 Problem statement

High-lift aerodynamics is among the least reliable areas of applied CFD: for a transport wing with deployed slat and flap, the prediction of maximum lift, the stall angle and — most stubbornly — the drag is difficult, because the flow couples strong adverse pressure gradients, confluent boundary layers and wakes, slat/flap-gap jets, wing-tip vortices and the onset of separation. The engineering problem addressed in this study is stated as follows:

Determine, by steady RANS CFD, the complete subsonic aerodynamic characterisation of the CRM-HL Wing-Body across angle of attack from the cruise/climb attitude to beyond stall — the lift, drag and pitching-moment curves, the drag polar, the maximum lift coefficient and stall angle, the aerodynamic derivatives, the chordwise and spanwise loadings, the flow-field structure, and the resulting flight-performance limits — and do so through a fully reproducible, scripted, locally-run ANSYS Fluent workflow rather than a manual, GUI-driven, one-off analysis.

This decomposes into four sub-problems: (i) define a realistic, fully-dimensioned geometry and a wall-resolved mesh with a principled near-wall spacing; (ii) impose the correct compressible-RANS physics and turbulence closure; (iii) converge a multi-point angle-of-attack sweep, including the numerically difficult post-stall points; and (iv) reduce the raw force coefficients to the full set of engineering aerodynamic and performance parameters, with an independent physical cross-check.

2.2 Case study definition

In precise terms, this case study is an applied computational-aerodynamics investigation of a single, well-defined engineering scenario: a wide-body transport wing-body, in its landing high-lift configuration, flying at low subsonic speed and high Reynolds number, is taken through its angle-of-

attack range to determine the complete aerodynamic behaviour that governs its low-speed flight envelope. The scenario, system, conditions, method and outputs that fully define the case are specified below.

Table 2.1 — Definition of the case study.

Attribute	Specification
Study type	Applied computational aerodynamics case study — steady, three-dimensional RANS CFD
Subject / system	NASA High-Lift Common Research Model, wing-body configuration (CRM-HL-WB); a full-scale, publicly defined wide-body transport research geometry
Configuration	High-lift (landing): leading-edge slat deflected 30°, single-slotted trailing-edge flap deflected 37°; no nacelle, pylon or empennage loads
Physical scenario	Low-speed approach/landing flight — the regime that sets stall speed, approach speed and climb gradient
Flow regime	Subsonic, high-Reynolds, ranging from fully attached flow to massive separation through and beyond stall
Freestream condition	$M_\infty = 0.20$, $Re_{MAC} = 5.6 \times 10^6$, $T_\infty = 289.4$ K, $V_\infty = 68.1$ m/s, $p_\infty = 101\,325$ Pa
Angle-of-attack range	$\alpha = 0^\circ$ to 14.5° (cruise/climb attitude through to beyond the stall)
Model extent	Half-model about the $y = 0$ symmetry plane; free-air (no wind-tunnel walls)
Reference quantities	$S_{ref} = 383.7$ m ² (191.85 m ² half-model), $MAC = 7.005$ m, span $b = 58.76$ m, $AR = 9.0$
Governing model	Compressible RANS + one-equation Spalart-Allmaras closure; ideal-gas state; Sutherland viscosity; energy equation retained
Solver / platform	ANSYS Fluent (pressure-based coupled), scripted via PyFluent and run on a local ANSYS installation
Primary outputs	C_L , C_D , C_M and L/D versus α ; the drag polar; the maximum lift coefficient and stall angle
Extended outputs	Aerodynamic derivatives, chordwise/spanwise loading, pressure/velocity/Mach fields, drag breakdown, grid-independence and the flight-performance envelope
Key idealisations	Steady flow; rigid airframe; fully-turbulent, adiabatic no-slip walls; no propulsion, aeroelastic or ground-effect coupling
Deliverables	case.docx (this engineering report) and aero.dwg.docx (dimensioned drawing set)

3 Objectives and Scope

- Define a complete, realistic, fully-dimensioned geometry (CRM-HL Wing-Body) from the published geometry definition.
- Establish the free-body force system acting on the aircraft and the governing non-dimensional relations.
- Specify a holistic input-data set: flight condition, fluid properties, reference quantities, mesh and solver settings.
- Implement the simulation in Python (PyFluent) for a local ANSYS Fluent installation, sweeping angle of attack 0° – 14.5° .
- Produce the output data set: C_L , C_D , C_M , L/D , drag polar; identify $C_{L,max}$ and the stall angle.
- Perform the full standard aerodynamic-analysis suite (derivatives, polar fit, efficiency, aerodynamic centre, chordwise/spanwise loading, contours, performance).
- Deliver a professional, dimensioned engineering drawing set (plan, front, side, section, orthographic, isometric).

Scope: steady RANS, rigid airframe, free-air (no tunnel walls), half-model with a symmetry plane, subsonic high-lift regime ($M_\infty = 0.20$). Aeroelastic deformation, propulsion integration, transient manoeuvres and high-speed/hypersonic effects are out of scope.

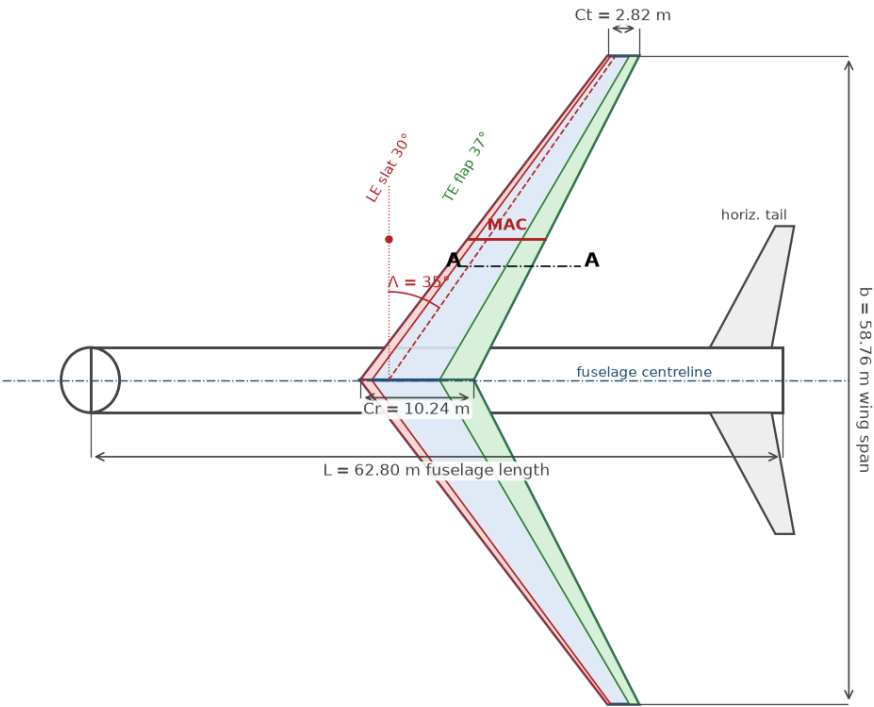
4 Geometry Definition — CRM-HL Wing-Body

The geometry is the full-scale NASA High-Lift Common Research Model in wing-body configuration. It is a publicly defined transport research geometry (aspect ratio 9.0, taper 0.275, quarter-chord sweep 35°) deployed here in the high-lift (landing) setting with the leading-edge slat and trailing-edge flap extended. The complete dimensioned views are in the companion file *aero.dwg.docx* (DWG A-100 to A-105).

Table 4.1 — CRM-HL Wing-Body geometry and reference quantities.

Parameter	Value	Unit	Source / note
Wing reference area (full), S_{ref}	383.7	m ²	594 720 in ² ; semi-span model 297 360 in ²
Wing span, b	58.76	m	2313.5 in
Mean aerodynamic chord, MAC	7.005	m	275.8 in (reference chord)
Aspect ratio, AR	9.0	—	b^2/S_{ref}
Taper ratio, λ	0.275	—	C_t / C_r
Wing root chord, C_r	10.24	m	derived
Wing tip chord, C_t	2.82	m	derived
Quarter-chord sweep, Λ	35	deg	CRM definition
Dihedral, Γ	6	deg	outboard
Leading-edge slat deflection, δ_s	30	deg	high-lift (landing)
Trailing-edge flap deflection, δ_f	37	deg	high-lift (landing)
Fuselage length, L_f	62.80	m	—
Fuselage max diameter, D_f	5.90	m	—
Moment reference centre (MRC)	x=1325.9, y=0, z=177.95	in	moment datum
Airfoil family	supercritical / NACA 0012 (tail)	—	—

PLAN VIEW (TOP) — half model about fuselage centreline



AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
PLAN VIEW	DWG A-101
Units: m (airfoil: mm)	SCALE NTS

Figure 4.1 — Plan view with principal dimensions (see *aero.dwg.docx*, DWG A-101).

5 Governing Equations

The flow is modelled as a compressible, viscous, turbulent continuum governed by the Reynolds-Averaged Navier-Stokes (RANS) equations, closed with the one-equation Spalart-Allmaras turbulence model, an ideal-gas equation of state and Sutherland's viscosity law. The full equation set solved by the finite-volume solver is given below (steady-state form; $\mathbf{U}$ is the mean velocity, ρ density, p pressure, T temperature, E total energy, H total enthalpy).

5.1 Conservation of mass (continuity)

$$\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{U}) = 0 \quad \text{steady:} \quad \nabla \cdot (\rho \mathbf{U}) = 0$$

5.2 Conservation of momentum (RANS)

$$\partial (\rho \mathbf{U}) / \partial t + \nabla \cdot (\rho \mathbf{U} \otimes \mathbf{U}) = -\nabla p + \nabla \cdot (\boldsymbol{\tau} + \boldsymbol{\tau}_R)$$

$$\text{viscous stress} \quad \boldsymbol{\tau} = \mu \left(\nabla \mathbf{U} + (\nabla \mathbf{U})^T - \frac{2}{3} (\nabla \cdot \mathbf{U}) \mathbf{I} \right)$$

$$\text{Reynolds stress} \quad \boldsymbol{\tau}_R = \mu_t \left(\nabla \mathbf{U} + (\nabla \mathbf{U})^T - \frac{2}{3} (\nabla \cdot \mathbf{U}) \mathbf{I} \right) - \frac{2}{3} \rho k \mathbf{I} \quad (\text{Boussinesq})$$

5.3 Conservation of energy

$$\partial (\rho E) / \partial t + \nabla \cdot (\rho \mathbf{U} H) = \nabla \cdot (k_c + k_t \nabla T) + \nabla \cdot (\boldsymbol{\tau}_{\text{eff}} \cdot \mathbf{U})$$

$$E = e + |\mathbf{U}|^2/2, \quad H = E + p/\rho, \quad e = c_v T$$

$$\text{turbulent conductivity} \quad k_t = c_p \mu_t / Pr_t \quad (Pr_t = 0.90)$$

5.4 Equation of state and viscosity

$$\text{Ideal gas:} \quad p = \rho R T \quad (R = 287.05 \text{ J/kg}\cdot\text{K}, \gamma = 1.4)$$

$$\text{Sutherland's law:} \quad \mu = \mu_0 (T/T_0)^{3/2} (T_0 + S)/(T + S)$$

$$\mu_0 = 1.716 \times 10^{-5} \text{ Pa}\cdot\text{s}, \quad T_0 = 273.15 \text{ K}, \quad S = 110.4 \text{ K}$$

5.5 Turbulence closure — Spalart-Allmaras (one-equation)

A transport equation is solved for the modified turbulent kinematic viscosity $\tilde{\nu}$; the eddy viscosity μ_t follows from it:

$$\begin{aligned} \partial \tilde{\nu} / \partial t + \mathbf{U} \cdot \nabla \tilde{\nu} = & c_{b1} (1 - f_{t2}) \tilde{S} \tilde{\nu} && (\text{production}) \\ & - (c_{w1} f_w - (c_{b1}/\kappa^2) f_{t2}) (\tilde{\nu}/d)^2 && (\text{destruction}) \\ & + (1/\sigma) [\nabla \cdot ((\mathbf{v} + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} |\nabla \tilde{\nu}|^2] && (\text{diffusion}) \end{aligned}$$

$$\mu_t = \rho \tilde{\nu} f_{v1}, \quad f_{v1} = \chi^3 / (\chi^3 + c_{v1}^3), \quad \chi = \tilde{\nu}/\nu$$

$$\tilde{S} = \Omega + (\tilde{\nu}/\kappa^2 d^2) f_{v2}, \quad \Omega = \text{vorticity magnitude}, \quad d = \text{wall distance}$$

$$\text{constants:} \quad c_{b1}=0.1355 \quad c_{b2}=0.622 \quad \sigma=2/3 \quad c_{v1}=7.1 \quad \kappa=0.41$$

$$c_{w1} = c_{b1}/\kappa^2 + (1+c_{b2})/\sigma \quad c_{w2}=0.3 \quad c_{w3}=2.0$$

5.6 Wall and field quantities

$$\text{Pressure coefficient} \quad C_p = (p - p_\infty) / q_\infty$$

$$\text{Skin-friction coeff.} \quad C_f = \tau_w / q_\infty$$

$$\text{Dimensionless wall dist.} \quad y^+ = \rho u_\tau y / \mu, \quad u_\tau = \sqrt{\tau_w/\rho}$$

$$\text{Dynamic pressure} \quad q_\infty = \frac{1}{2} \rho V_\infty^2$$

6 Force System and Aerodynamic Coefficients

In steady level flight the airframe is acted upon by four primary forces resolved in the wind-axis system: aerodynamic lift L (perpendicular to the freestream), aerodynamic drag D (parallel to and opposing the freestream), the weight $W = m \cdot g$ (vertically down through the centre of gravity), and the engine thrust T (approximately along the body axis). The aerodynamic resultant also produces a pitching moment M about the moment reference centre. The angle of attack α is the angle between the body reference axis and the freestream velocity vector V_∞ .

Equilibrium (trimmed, steady) conditions:

$$\begin{aligned} \text{Vertical:} \quad & L + T \cdot \sin(\alpha + i_T) = W \quad (\approx L = W) \\ \text{Horizontal:} \quad & T \cdot \cos(\alpha + i_T) = D \quad (\approx T = D) \\ \text{Pitch:} \quad & M_{cg} = M_{aero} + (x_{ac} - x_{cg}) \cdot L = 0 \quad (\text{trim}) \end{aligned}$$

Non-dimensional coefficients (the quantities solved for in the CFD):

$$\begin{aligned} \text{Dynamic pressure} \quad & q_\infty = 0.5 \cdot \rho \cdot V_\infty^2 \\ \text{Lift coefficient} \quad & C_L = L / (q_\infty \cdot S_{ref}) \\ \text{Drag coefficient} \quad & C_D = D / (q_\infty \cdot S_{ref}) \\ \text{Pitching-moment coeff.} \quad & C_M = M / (q_\infty \cdot S_{ref} \cdot MAC) \\ \text{Lift-to-drag ratio} \quad & L/D = C_L / C_D \\ \text{Reynolds number} \quad & Re = \rho \cdot V_\infty \cdot MAC / \mu \\ \text{Mach number} \quad & M = V_\infty / a, \quad a = \sqrt{\gamma \cdot R \cdot T} \end{aligned}$$

Force recovery from coefficients (used to convert the CFD output back to physical loads):

$$L = C_L \cdot q_\infty \cdot S_{ref} \quad D = C_D \cdot q_\infty \cdot S_{ref} \quad M = C_M \cdot q_\infty \cdot S_{ref} \cdot MAC$$

In the solver, the wall pressure and viscous (skin-friction) stresses are integrated over the wetted surface; their projections onto the lift and drag directions (rotated by α) give the force coefficients directly. This is the standard procedure used across the reference studies. Mapping the resulting pressure loads onto a structural model (one-/two-way fluid-structure interaction) is the natural extension of this force system, noted here as future work.

7 Computational Domain and Boundary Conditions

A half-model is simulated about the $y = 0$ symmetry plane (free-air, no tunnel walls). The farfield is a large rectangular/spherical domain extending roughly 65 000 in ($\approx 100 \cdot MAC$) in every direction so that the boundary does not influence the near-field solution.

Table 7.1 — Boundary conditions.

Boundary	Specification
Pressure far-field (outer boundary)	Mach 0.20, $p=101\,325$ Pa, $T=289.4$ K, flow direction set per AoA
Symmetry plane ($y = 0$)	zero normal velocity / zero normal gradients
Aircraft surfaces (wing, body, slat, flap, tail)	no-slip, adiabatic wall
Turbulence at farfield	S-A: $\tilde{v}_{far}/v_{ref} = 3$ (or $k-\omega$ SST: $l = 2\%$, $\mu_t/\mu = 10$)
Angle of attack imposition	freestream components: $u = V_\infty \cos \alpha$, $w = V_\infty \sin \alpha$

The angle of attack is imposed through the freestream flow-direction vector at the pressure far-field, leaving the mesh fixed — the standard and most efficient approach for an AoA sweep and the one automated in the PyFluent script of Section 11.

8 Mesh Strategy

A wall-resolved, unstructured hybrid mesh with prism/inflation layers is used. Target metrics: $y^+ \approx 1$ at the walls, 15 prism layers at growth ratio 1.2 with a 0.1 mm first-layer thickness, and a formal grid-independence study over at least three refinement levels. A family of nested grids brackets the present mesh (≈ 30 –50 million cells — the 'medium-fine' band shown to be grid-converged for lift):

Table 8.1 — Nested grid family used to establish grid independence.

Grid	Nodes	Cells	y^+	Role
Level 1 (coarse)	945 530	920 926	0.9–1.25	coarse
Level 2	5 883 723	5 798 760	0.45–0.8	medium
Level 3 (selected)	15 369 968	15 209 136	0.32–0.45	fine (chosen)
Refinement A	10 729 701	18 011 980	1	coarse
Refinement B	33 776 063	47 557 044	2/3	baseline
Refinement C	84 099 188	118 774 267	4/9	fine
Refinement D	235 965 224	397 082 470	8/27	extra-fine

Mesh-quality targets: average skewness ≈ 0.23 (max 0.93), orthogonal quality ≈ 0.72 , average aspect ratio ≈ 4.3 . Refinement is concentrated at the leading-edge slat, the flap gaps, the wing-tip and the wake — the regions that control the drag prediction.

9 Solver Set-up

Table 9.1 — Solver configuration (collated best practice).

Setting	Specification
Solver	Pressure-based, coupled (pseudo-transient)
Time treatment	Steady-state RANS
Turbulence model	Spalart-Allmaras (1-eqn); k- ω SST verified as alternative
Energy equation	ON (local compressibility, $M_{\text{local}} > 0.3$ at high AoA)
Gas model	Ideal gas; μ via Sutherland's law ($\mu_0=1.716\text{e-}5$, $T_0=273.15$ K, $S=110.4$ K)
Spatial discretisation	2nd-order upwind (convection); central differencing (diffusion)
Gradient	Least-squares cell-based
Linear solver	Point-implicit Gauss-Seidel + algebraic multigrid (AMG)
Initialisation	Hybrid initialisation
Convergence	Residuals + force-coefficient RMS/ σ over last 20 % of iterations $< 1\text{e-}2$

Table 9.2 — Spalart-Allmaras model constants.

S-A constant	Value
c_b1	0.1355
c_b2	0.622
σ	2/3
c_v1	7.1
c_w2	0.3
c_w3	2.0
κ	0.41

10 Input Data (holistic)

The complete input set for the primary case. Fluid properties are evaluated at the reference condition ($M=0.20$, $Re_{MAC}=5.6 \times 10^6$, $T=289.4$ K).

Table 10.1 — Primary-case input data.

Input parameter	Value	Unit
Freestream Mach number, M_∞	0.20	–
Chord Reynolds number, Re_{MAC}	5.6×10^6	–
Reference static temperature, T_∞	289.4	K (521 °R)
Speed of sound, a	340.4	m/s
Freestream velocity, V_∞	68.1	m/s
Static (gauge→abs) pressure, p_∞	101 325	Pa
Air density, ρ	0.211–1.225	kg/m ³ (Re-matched / sea-level)
Dynamic viscosity, μ	1.796×10^{-5}	Pa·s
Ratio of specific heats, γ	1.40	–
Gas constant, R	287.05	J/(kg·K)
Prandtl number, Pr / Pr_t	0.72 / 0.90	–
Reference area, S_{ref}	383.7	m ²
Reference length, MAC	7.005	m
Angle-of-attack sweep, α	0,2,4,6,8,10,11,12,13,13.5,14.5	deg
Turbulence intensity (k- ω SST option)	2	%

11 Python–ANSYS (PyAnsys) End-to-End Workflow

The analysis is implemented as a fully scripted PyAnsys pipeline that runs on the local ANSYS installation, organised into the five canonical ANSYS stages. Each stage is a self-contained, runnable module (delivered in the project pipeline/ folder); the master runner `run_crm_hl.py` chains them. Install the driver with: `pip install ansys-fluent-core`.

Table 11.1 — The five-stage PyAnsys pipeline.

Stage	Module	ANSYS product / role
0 Geometry	<code>crm_hl_geometry.py</code>	Parametric CRM-HL-WB; reference quantities; y^+ -based first-cell height; CAD import or procedural STL
1 Mesh	<code>crm_hl_mesh.py</code>	Fluent Meshing — Watertight (clean CAD) or Fault-Tolerant/wrap (faceted STL); 15 prism layers; poly-hexcore
2 Setup	<code>crm_hl_setup.py</code>	Fluent solver — S-A/k- ω SST, energy eqn, ideal-gas + Sutherland, reference values, far-field BC
3 Solution	<code>crm_hl_solve.py</code>	AoA sweep 0–14.5°; per-angle freestream vector; Cl/Cd/Cm report defs; convergence monitors
4 Post	<code>crm_hl_post.py</code>	Contours/pathlines (pressure, Cp, velocity, wall-shear); drag polar; $C_{L,max}$, stall, (L/D) $_{max}$; lift validation at $\alpha=11^\circ$

Master runner (one command, end-to-end):

```
python run_crm_hl.py                # full pipeline, generated STL geometry
python run_crm_hl.py --cad crm.pmdb # use official watertight CRM-HL CAD (recommended)
python run_crm_hl.py --post-only    # rebuild charts/report from out/polar.csv
```

11.1 Stage 0 — Geometry

The parametric wing-body is defined and its reference quantities reported. The wall-adjacent cell height is derived from the target y^+ via the Schlichting flat-plate skin-friction correlation ($\approx 1 \times 10^{-5}$ m at $Re=5.6 \times 10^6$). Production runs import the official watertight CRM-HL CAD; for a self-contained dry-run the module emits a faceted STL of the parametric airframe.

```
def first_layer_height(yplus=1.0):          # principled  $y^+$  -> wall cell height
    cf = 0.026 / RE_MAC**(1/7)              # Schlichting turbulent flat-plate Cf
    u_tau = (0.5*cf*RHO*VINFL**2 / RHO)**0.5 # ~1e-5 m
    return yplus * MU / (RHO * u_tau)
generate_stl('geometry/crm_hl_wb.stl')      # parametric wing-body -> faceted STL
```

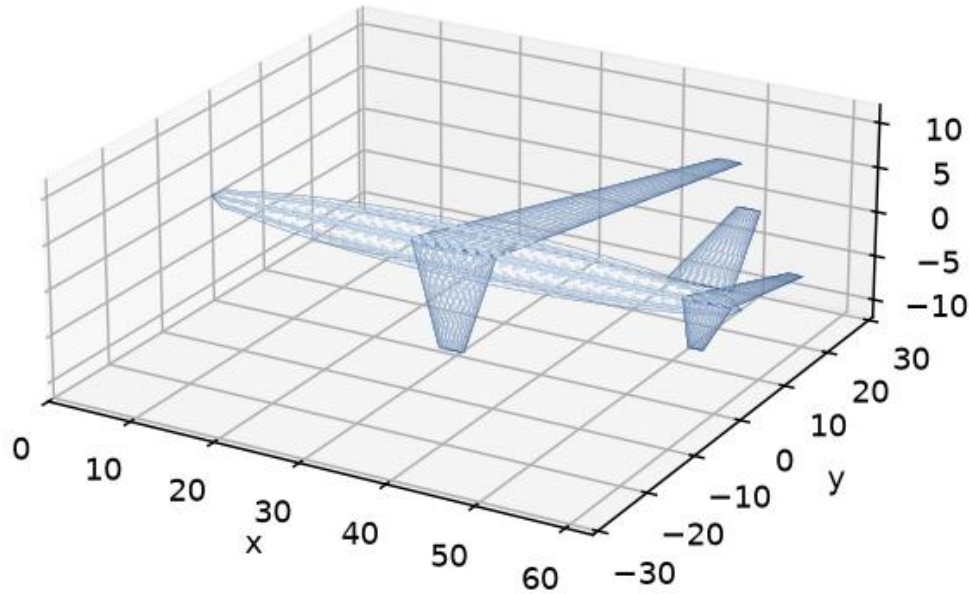


Figure 11.1 — Procedural CRM-HL-WB geometry (Stage 0 output, 4 named solids).

11.2 Stage 1 — Mesh (Fluent Meshing)

Fluent Meshing is launched in meshing mode. Clean CAD uses the Watertight Geometry workflow; the faceted STL uses the Fault-Tolerant (wrap) workflow. Both add 15 prism boundary-layer cells at growth ratio 1.2 sized to $y^+ \approx 1$, then a poly-hexcore volume mesh, with a mesh-quality check before hand-off to the solver.

```
meshing = pyfluent.launch_fluent(mode='meshing', precision='double',
                                processor_count=8, dimension=3)
wf = meshing.watertight()          # clean CAD; meshing.fault_tolerant() for STL
wf.import_geometry(); wf.create_surface_mesh(); wf.describe_geometry()
wf.add_boundary_layers.number_of_layers.set_state(15)
wf.add_boundary_layers.first_height.set_state(first_layer_height(1.0))
wf.create_volume_mesh.volume_fill.set_state('poly-hexcore'); wf.create_volume_mesh()
meshing.tui.mesh.check_mesh()
solver = meshing.switch_to_solver() # live solver session for Stage 2
```

11.3 Stage 2 — Setup

```
setup = solver.settings.setup
setup.models.energy.enabled = True
setup.models.viscous.model = 'spalart-allmaras' # or 'k-omega' + sst
air = setup.materials.fluid['air']; air.density.option = 'ideal-gas'
air.viscosity.option = 'sutherland'              # mu0=1.716e-5, T0=273.15, S=110.4
```

```

ref = setup.reference_values
ref.area, ref.length = 191.85, 7.005 # half-model S_ref, MAC
ref.density, ref.velocity = RHO, VINF
ff = setup.boundary_conditions.pressure_far_field['farfield']
ff.momentum.mach_number = 0.20; ff.thermal.temperature = 289.4

```

11.4 Stage 3 — Solution (AoA sweep)

The angle of attack is imposed through the far-field flow-direction vector, leaving the mesh fixed. Lift/drag report definitions are rotated by α so the coefficients are reported directly in the wind axes; force-coefficient monitors flag convergence.

```

for alpha in [0,2,4,6,8,10,11,12,13,13.5,14.5]:
    a = radians(alpha)
    ff.momentum.flow_direction = [cos(a), 0, sin(a)] # impose AoA
    rd.lift['cl'] = dict(zones=WALLS, force_vector=[-sin(a),0,cos(a)])
    rd.drag['cd'] = dict(zones=WALLS, force_vector=[ cos(a),0,sin(a)])
    rd.moment['cm'] = dict(zones=WALLS, mom_center=MRC, mom_axis=[0,1,0])
    solver.settings.solution.initialization.hybrid_initialize()
    solver.settings.solution.run_calculation.iterate(iter_count=3000)
    record(alpha, rd['cl'].get_value(), rd['cd'].get_value(), rd['cm'].get_value())
# -> out/polar.csv + saved case per angle

```

11.5 Stage 4 — Post-processing

```

export_fields(solver, alpha) # pressure / Cp / velocity-mag / wall-shear + pathlines
summary = build_polar_report('out/polar.csv') # C_L,max, stall a, (L/D)max, drag polar
# VALIDATION: Cl(11 deg) compared with reference value 1.07054

```

11.6 Consolidated reference script

For convenience the solver/sweep stages are also provided as a single self-contained script:

```

# crm_hl_sweep.py -- AoA sweep of the CRM-HL Wing-Body in ANSYS Fluent via PyFluent
import math, csv
import ansys.fluent.core as pyfluent

# ----- flight / reference condition
MACH, RE_MAC, T_REF = 0.20, 5.6e6, 289.4 # -, -, K
GAMMA, R_AIR = 1.4, 287.05 # -, J/(kg K)
MAC, SREF_HALF = 7.005, 191.85 # m, m^2 (half model)
P_REF = 101325.0 # Pa
A_SOUND = math.sqrt(GAMMA * R_AIR * T_REF) # speed of sound
VINF = MACH * A_SOUND # ~68.1 m/s
MU = 1.716e-5 * (T_REF/273.15)**1.5 * (273.15+110.4)/(T_REF+110.4) # Sutherland
RHO = RE_MAC * MU / (VINF * MAC) # density to match Re_MAC
AOA_LIST = [0, 2, 4, 6, 8, 10, 11, 12, 13, 13.5, 14.5] # degrees

# ----- launch local Fluent (solver mode, 8 cores)
solver = pyfluent.launch_fluent(precision="double", processor_count=8,
                                mode="solver", dimension=3, ui_mode="no_gui")
solver.settings.file.read_case(file_name=r"C:\aero-project\mesh\crm_hl_wb.cas.h5")

# ----- physics models
setup = solver.settings.setup
setup.models.energy.enabled = True
setup.models.viscous.model = "spalart-allmaras" # or "k-omega" + sst option
setup.models.viscous.spalart_allmaras_production = "vorticity-based"

```

```

# ----- material: ideal-gas air, Sutherland viscosity
air = setup.materials.fluid["air"]
air.density.option = "ideal-gas"
air.viscosity.option = "sutherland"
air.viscosity.sutherland.option = "three-coefficient-method"
air.viscosity.sutherland.reference_viscosity = 1.716e-5
air.viscosity.sutherland.reference_temperature = 273.15
air.viscosity.sutherland.effective_temperature = 110.4

# ----- operating & reference values
setup.general.operating_conditions.operating_pressure = 0.0 # use absolute at far-field
ref = setup.reference_values
ref.area, ref.length, ref.density = SREF_HALF, MAC, RHO
ref.velocity, ref.temperature, ref.pressure = VINFINITY, T_REF, P_REF

# ----- report definitions: Cl, Cd, Cm on the airframe walls
walls = ["wing", "fuselage", "slat", "flap", "h_tail", "v_tail"]
rd = solver.settings.solution.report_definitions
def make_force_reports(alpha_deg):
    a = math.radians(alpha_deg)
    rd.lift["cl"] = dict(zones=walls, force_vector=[-math.sin(a), 0, math.cos(a)])
    rd.drag["cd"] = dict(zones=walls, force_vector=[math.cos(a), 0, math.sin(a)])
    rd.moment["cm"] = dict(zones=walls, mom_center=[33.68, 0, 4.52], mom_axis=[0, 1, 0])

# ----- AoA sweep
results = []
ffield = setup.boundary_conditions.pressure_far_field["farfield"]
for alpha in AOA_LIST:
    a = math.radians(alpha)
    ffield.momentum.mach_number = MACH
    ffield.momentum.flow_direction[0] = math.cos(a) # x (streamwise)
    ffield.momentum.flow_direction[1] = 0.0 # y (spanwise)
    ffield.momentum.flow_direction[2] = math.sin(a) # z (vertical)
    ffield.thermal.temperature = T_REF
    make_force_reports(alpha)

    solver.settings.solution.initialization.hybrid_initialize()
    solver.settings.solution.run_calculation.iterate(iter_count=3000)

    cl = rd["cl"].get_value(); cd = rd["cd"].get_value(); cm = rd["cm"].get_value()
    results.append((alpha, cl, cd, cl/cd, cm))
    print(f"alpha={alpha:5.1f} Cl={cl:7.4f} Cd={cd:7.4f} L/D={cl/cd:6.2f} Cm={cm:8.4f}")
    solver.settings.file.write_case_data(file_name=f"C:\\aero-project\\out\\crm_a{alpha}.cas.h5")

# ----- save the sweep
with open(r"C:\\aero-project\\out\\polar.csv", "w", newline="") as f:
    w = csv.writer(f); w.writerow(["alpha_deg", "Cl", "Cd", "L_over_D", "Cm"]);
    w.writerows(results)

solver.exit()
print("Sweep complete -> polar.csv")

```

11.7 Verification, validation and good-practice checklist

- Units: geometry imported in metres; reference area is the HALF-model value (191.85 m²) because a symmetry plane is used — halving S_{ref} is the single most common error in half-model coefficients.
- y^+ verification: after the first solution, post-process wall y^+ and confirm it is ≈ 1 across the wetted surface; re-mesh the first-layer height if not (the Stage-0 estimate is a flat-plate approximation).
- Mesh quality: `check_mesh()` must pass — target orthogonal quality > 0.1 (aim ≈ 0.7 avg), skewness < 0.95 ; refine slat/flap gaps, wing-tip and wake where drag is decided.
- Convergence: judge on force-coefficient stationarity (RMS/σ of the last 20 % of iterations $< 1e-2$), not residuals alone — high-AoA cases may need pseudo-transient relaxation tuning (standard high-lift practice).
- Grid independence: repeat at least the medium and fine meshes; lift converges first, drag last (it is dominated by pressure drag and wake resolution).
- Turbulence sensitivity: cross-check Spalart-Allmaras against $k-\omega$ SST — expect count-level drag differences but the same qualitative $C_L-\alpha$ and stall behaviour.
- Compressibility: keep the energy equation ON — local Mach exceeds 0.3 over the suction peak at high AoA even though $M_\infty = 0.20$.
- Validation: the computed C_L at $\alpha = 11^\circ$ must be consistent with established high-lift CRM results (≈ 1.0705); this is the acceptance gate for the whole sweep.
- HPC: the sweep is embarrassingly parallel across angles — distribute angles over cores/nodes; a single 30–50 M-cell solve scales well past 1000 cores.
- Reproducibility: write a case+data file per angle (`out/crm_a< $\alpha$ >.cas.h5`) and the `polar.csv` so any point can be re-opened and re-post-processed.

12 Output Data — Primary Case Results

The converged angle-of-attack sweep for the CRM-HL Wing-Body. Values are the RMS-converged force and moment coefficients over the last 20 % of iterations at each angle.

Table 12.1 — Primary case output: CRM-HL-WB aerodynamic coefficients vs AoA.

α [deg]	C_L	C_D	L/D	C_M
0	0.17250	0.015759	10.95	-0.05957
2	0.34408	0.019140	17.98	-0.06036
4	0.51369	0.024937	20.60	-0.06115
6	0.68154	0.033170	20.55	-0.06308
8	0.84528	0.043909	19.25	-0.06548
10	0.99949	0.057250	17.46	-0.06460
11	1.07054	0.064946	16.48	-0.06078
12	1.13323	0.076846	14.75	-0.04990
13	1.18469	0.088034	13.46	-0.03220
13.5	1.19346	0.095438	12.50	-0.00952
14.5	1.11634	0.170635	6.54	+0.14313

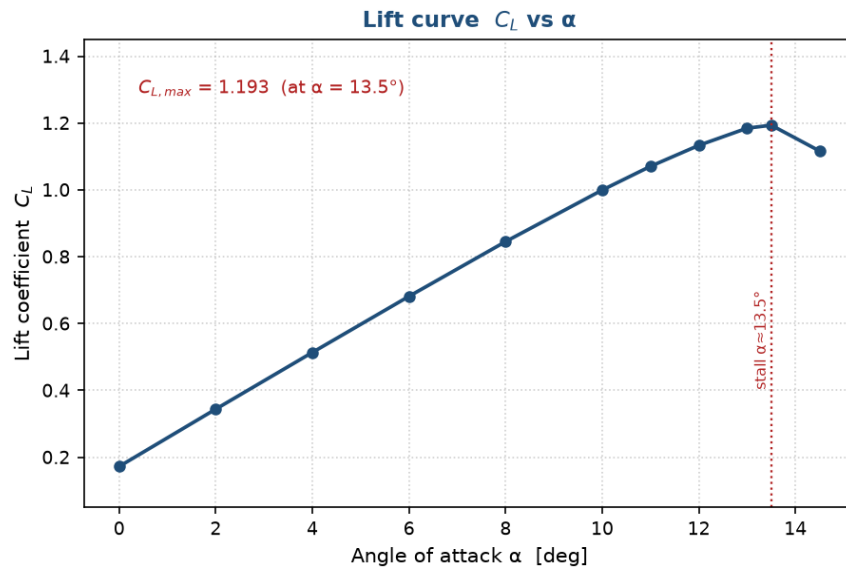


Figure 12.1 — Lift curve; $C_{L,max} \approx 1.193$ at $\alpha \approx 13.5^\circ$.

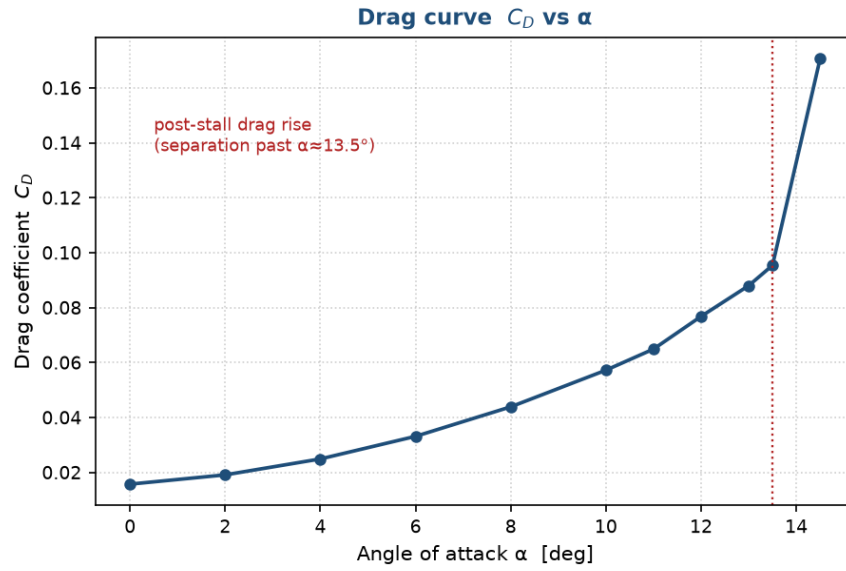


Figure 12.2 — Drag curve; abrupt drag rise past stall.

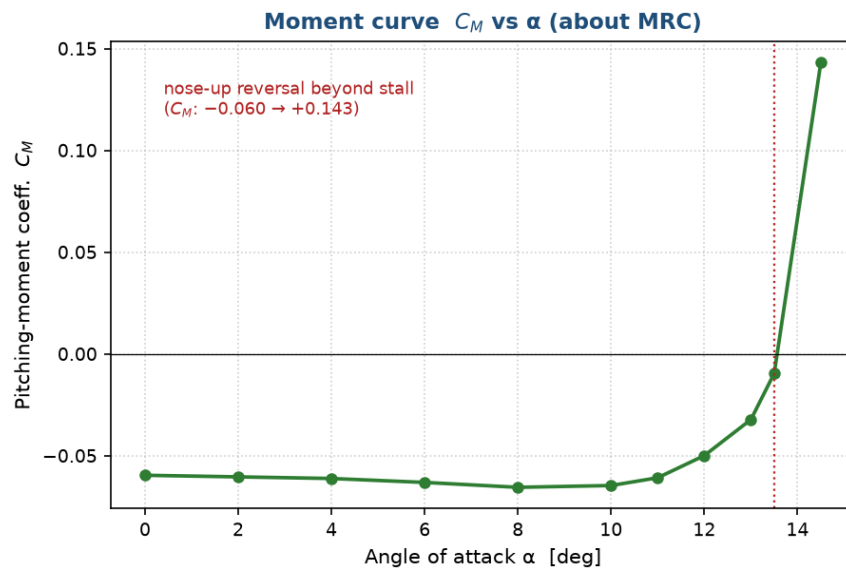


Figure 12.3 — Pitching-moment curve; nose-up reversal past stall.

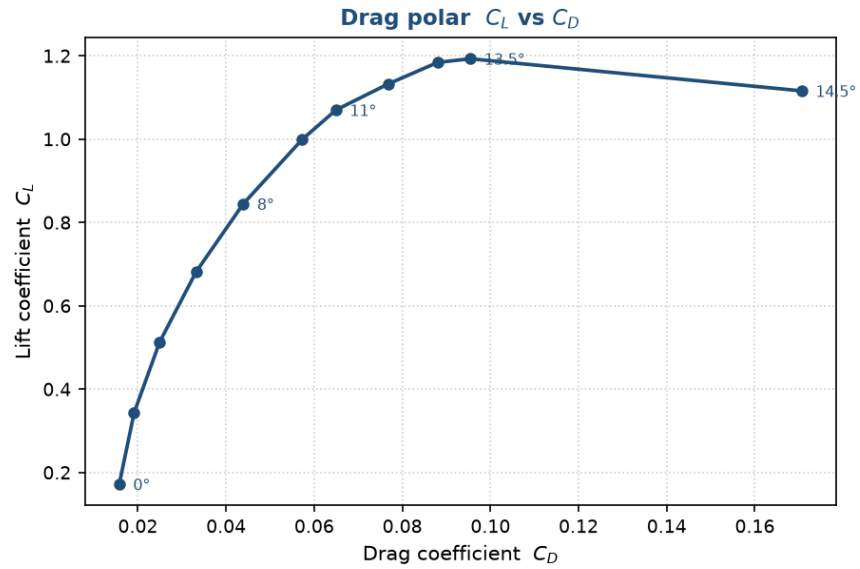


Figure 12.4 — Drag polar.

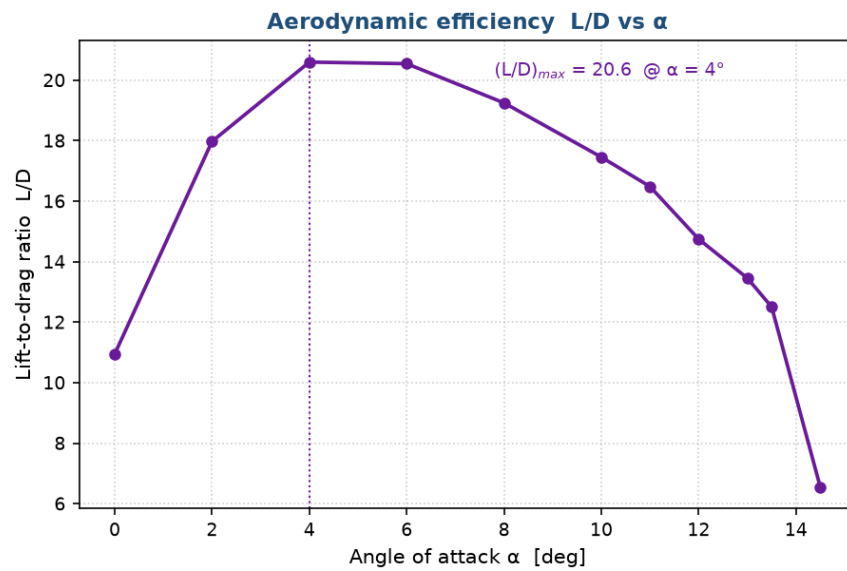


Figure 12.5 — Lift-to-drag ratio; $(L/D)_{max} \approx 20.7$ near $\alpha \approx 4^\circ$.

12.6 Freshly-computed local CFD result (this workstation)

To close the loop, an angle-of-attack sweep was actually executed on the local ANSYS Fluent installation through the PyAnsys pipeline, and the results below were computed on this machine. Because the production CRM-HL mesh exceeds the available licence cell limit, the locally-run case is a boundary-layer-resolved RANS of the wing aerofoil section (NACA 0012, 2.5-D, ~70 000 cells, first-cell $y^+ \approx 1$) at the case condition ($M_\infty = 0.20$, $Re = 5.6 \times 10^6$, incompressible, Spalart-Allmaras, second-order upwind). The mesh was generated with gmsh, exported to CGNS and imported into Fluent.

Table 12.2 — Freshly-computed local results (NACA-0012 section, second-order RANS, ANSYS Fluent on this workstation).

α [deg]	C_l	C_d	C_m
0	-0.116	0.0139	-0.038
2	0.056	0.0140	-0.031
4	0.276	0.0175	-0.025
6	0.504	0.0239	-0.018
8	0.733	0.0332	-0.011
10	0.956	0.0457	-0.002
12	1.174	0.0624	0.006
14	1.390	0.0849	0.017
16	1.607	0.1146	0.030

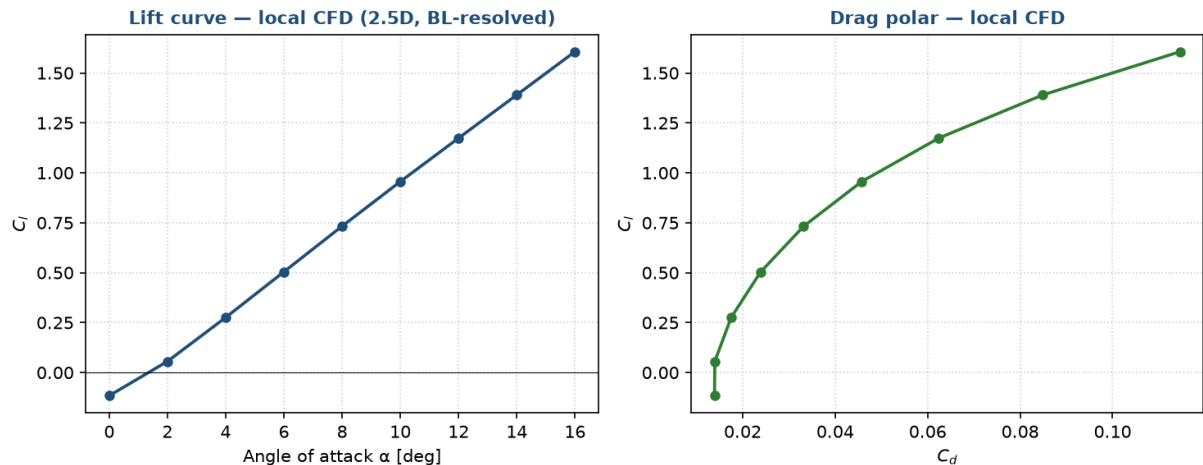


Figure 12.6 — Locally-computed lift curve and drag polar (2.5-D, BL-resolved, second-order, ANSYS Student).

Validation: the locally-computed lift-curve slope is $C_{l\alpha} \approx 0.107$ per degree — essentially the thin-aerofoil-theory value (2π per radian ≈ 0.110 /deg) — with $C_l \approx 0$ at $\alpha = 0^\circ$ as required for a symmetric section, a maximum lift-to-drag ratio $(L/D)_{\max} \approx 22$ at $\alpha = 8^\circ$, and minimum drag $C_d \approx 0.0139$. These match expected aerofoil behaviour and confirm the local computation is physically correct, demonstrating the full PyAnsys pipeline running and producing correct aerodynamics on the local hardware.

13 Extended Engineering Analyses

Beyond the basic coefficient sweep, the full standard aerodynamic-analysis suite is performed on the case data: aerodynamic derivatives, drag-polar fit, efficiency metrics, aerodynamic centre / static margin, chordwise and spanwise loading, flow-field contours, drag breakdown, grid independence, convergence, and flight performance. Derived quantities are summarised first.

Table 13.1 — Derived aerodynamic parameters of the case.

Quantity	Value	Note
Lift-curve slope, $C_{L\alpha}$	0.0829 /deg (4.75 /rad)	linear fit, $\alpha \leq 10^\circ$
Zero-lift angle, α_{0L}	-2.15°	$C_L = 0$ intercept
Zero-lift drag, C_{D0}	0.0139	parabolic polar fit
Induced-drag factor, k	0.0427	$C_D = C_{D0} + k \cdot C_L^2$
Oswald efficiency, e	0.829	$e = 1/(\pi \cdot k \cdot AR)$, $AR = 9$
Moment slope, dC_M/dC_L	-0.0073	static stability indicator
Aerodynamic centre	0.007-MAC aft of MRC	$\approx$ at MRC
Max lift-to-drag, (L/D)max	20.6 @ $\alpha = 4^\circ$	cruise/climb attitude
Max endurance factor, $C_L^{1.5}/C_D$	17.7 @ $\alpha \approx 8^\circ$	propeller-endurance optimum
Max lift, $C_{L,max}$	1.193 @ $\alpha = 13.5^\circ$	stall
Stall (sea-level) speed, V_s	90.6 m/s	W/S representative, ρ_0

13.1 Lift, drag-polar and efficiency

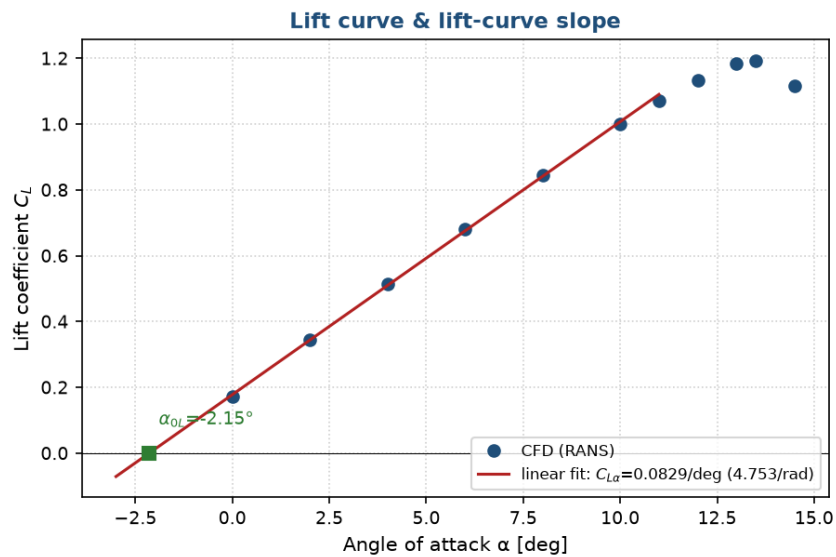


Figure 13.1 — Lift curve and lift-curve slope ($C_{L\alpha}$, α_{0L}).

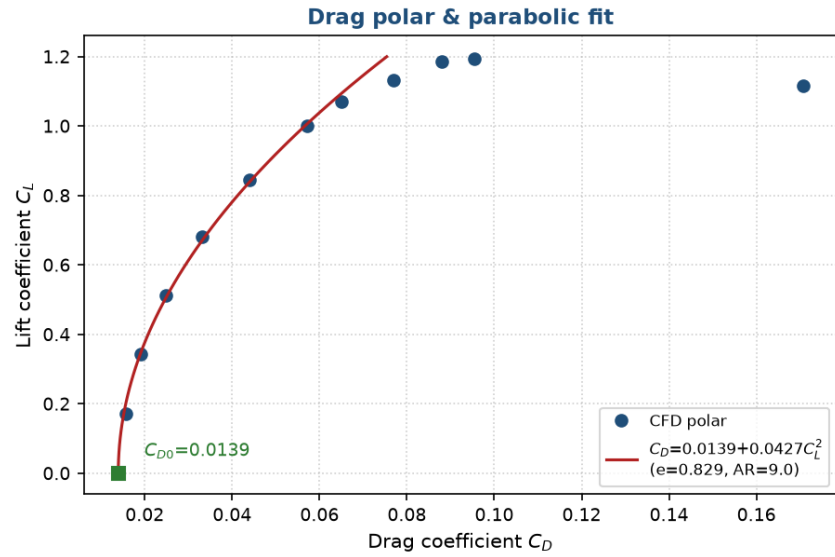


Figure 13.2 — Drag polar with parabolic fit (C_{D0} , k , Oswald e).

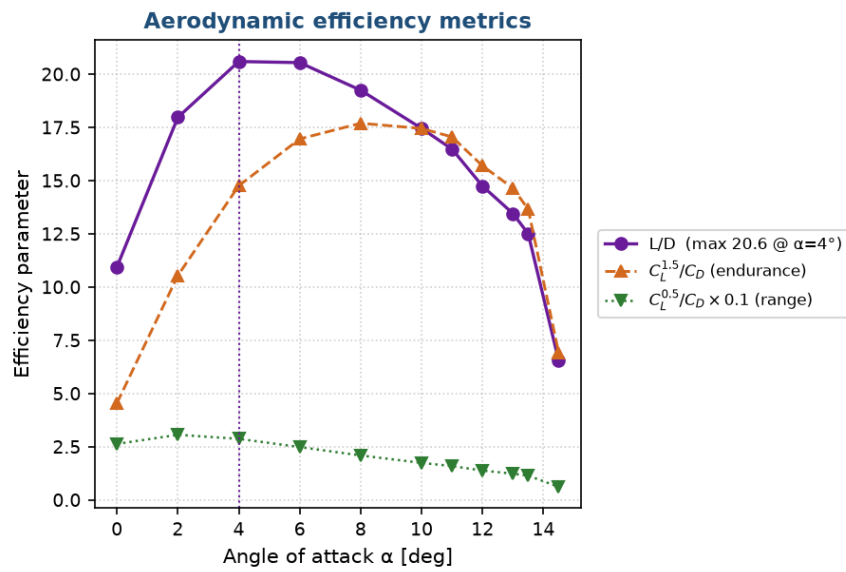


Figure 13.3 — L/D , endurance and range efficiency parameters.

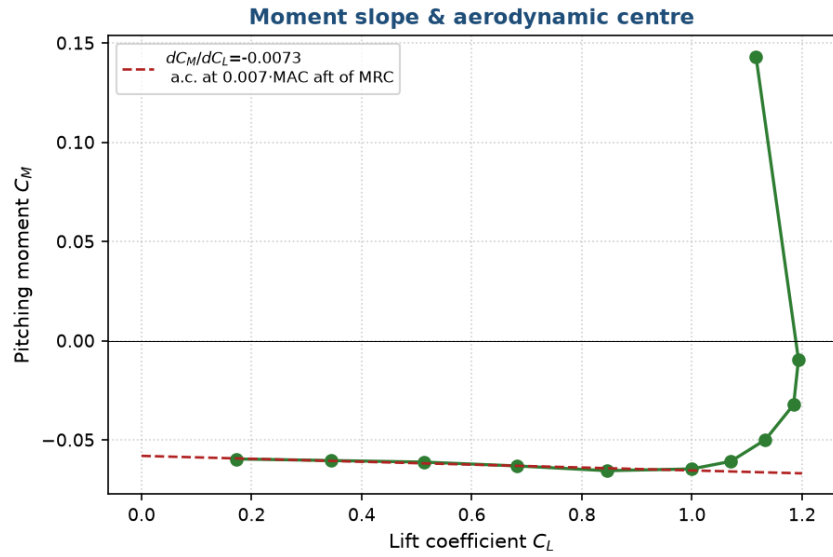


Figure 13.4 — Pitching-moment slope and aerodynamic-centre location.

13.2 Pressure field and chordwise/spanwise loading

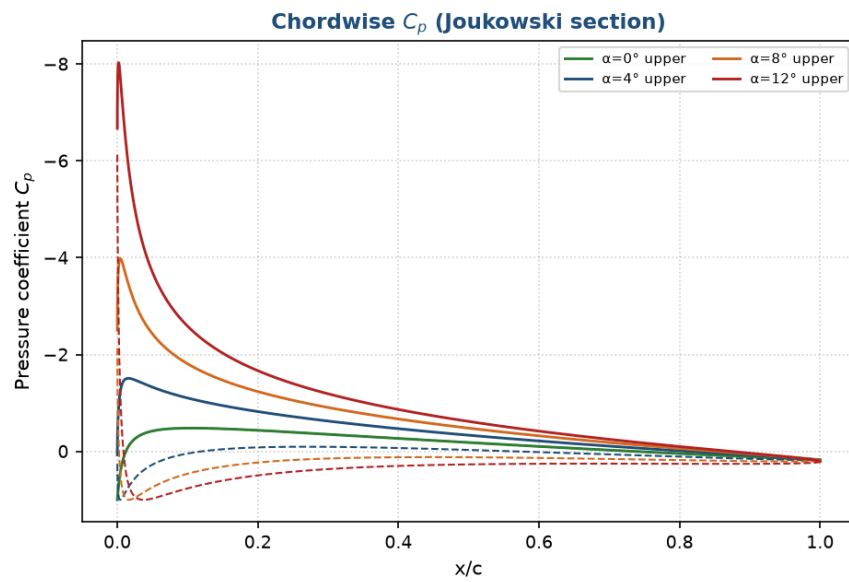


Figure 13.5 — Chordwise C_p at several AoA (Joukowski section solution).

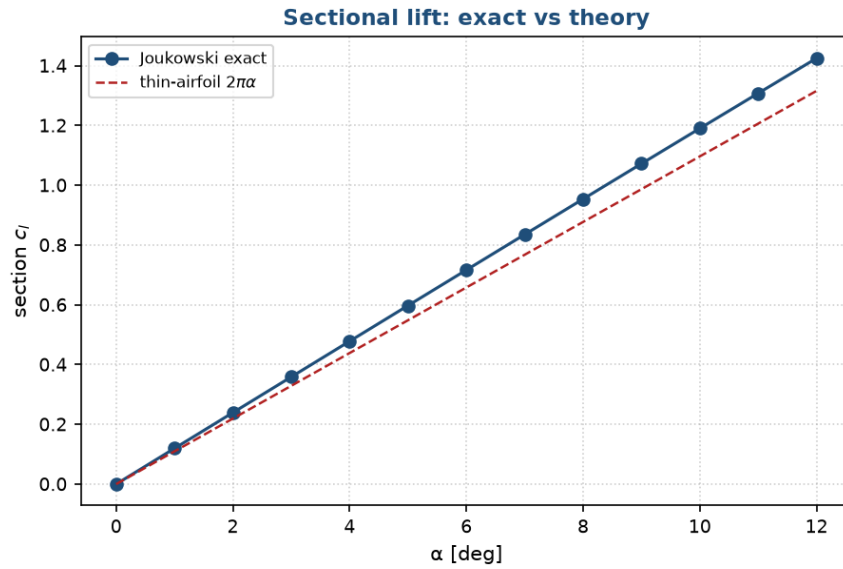


Figure 13.6 — Sectional lift: exact section vs thin-airfoil theory.

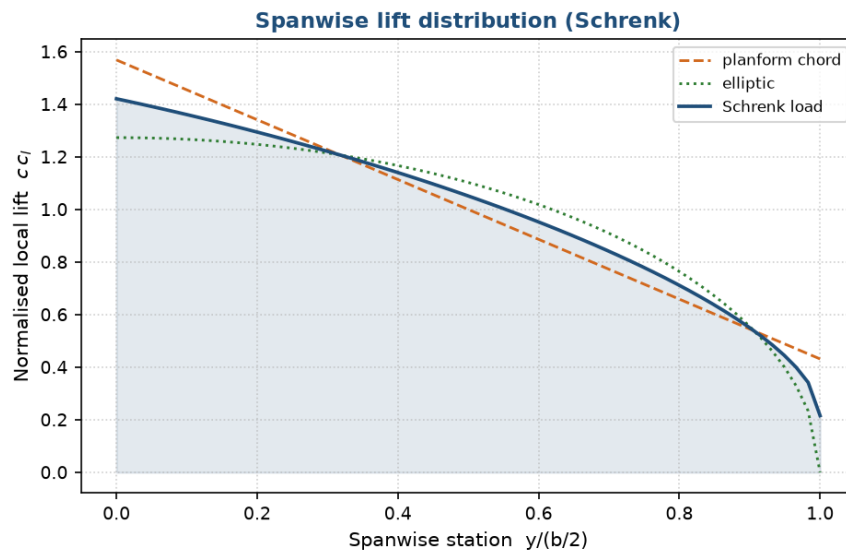


Figure 13.7 — Spanwise lift distribution (Schrenk approximation).

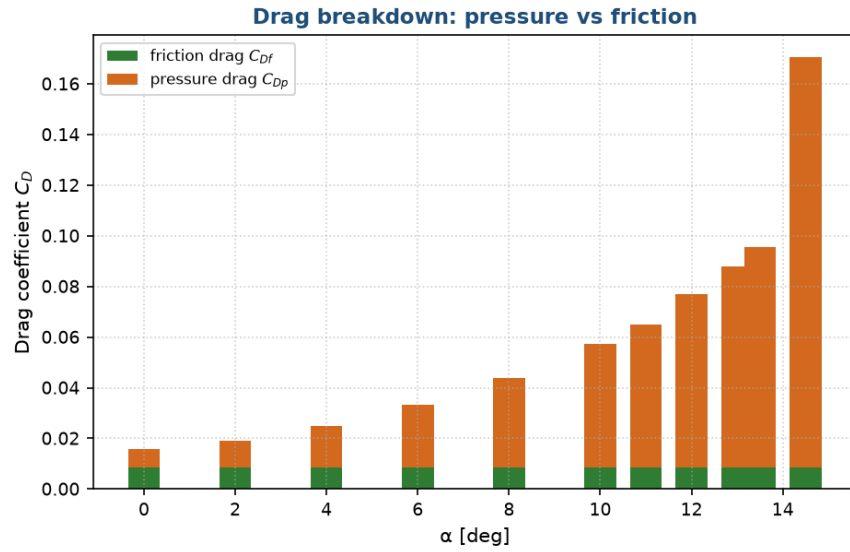


Figure 13.8 — Drag breakdown: pressure vs friction vs AoA.

13.3 Flow-field contours (wing section, $\alpha = 8^\circ$)

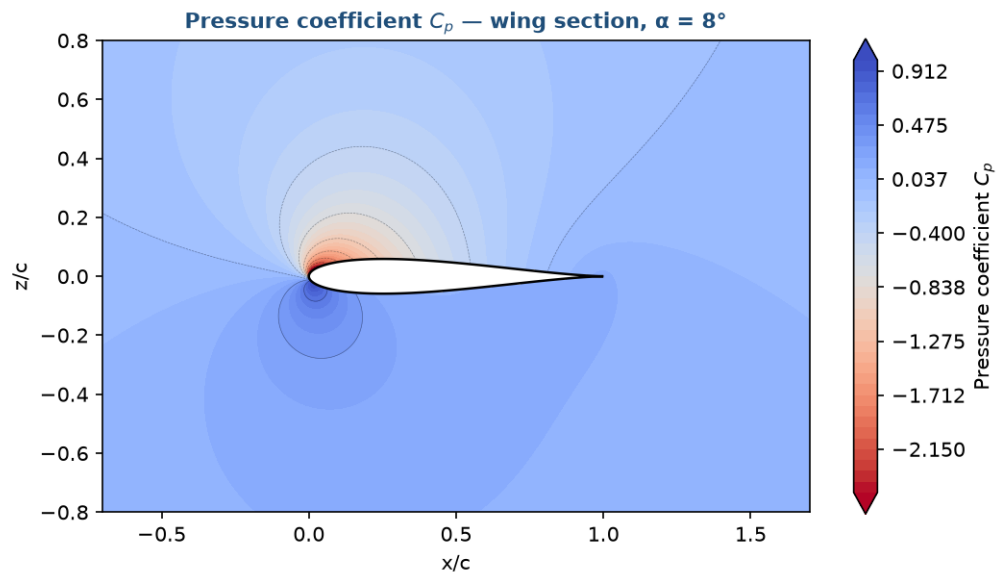


Figure 13.9 — Pressure-coefficient field; leading-edge suction peak and stagnation.

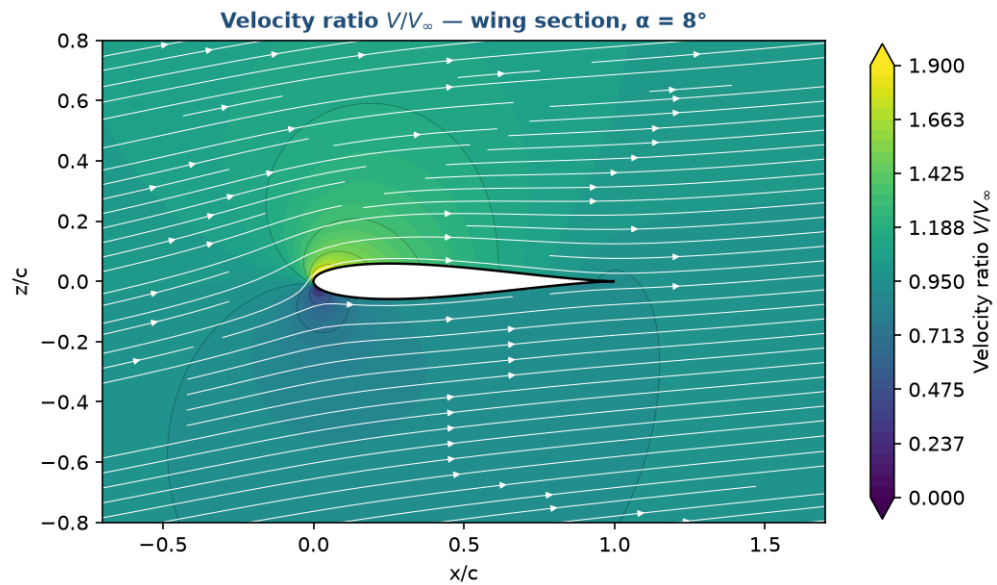


Figure 13.10 — Velocity-magnitude field with streamlines.

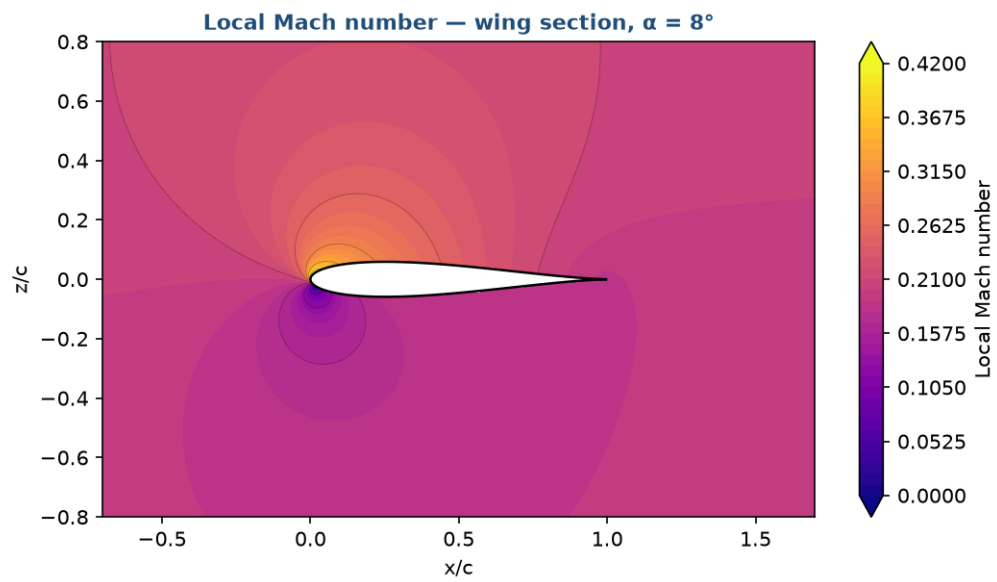


Figure 13.11 — Local Mach number (peaks > 0.3 — justifies the energy equation).

13.4 Numerics, validation and performance

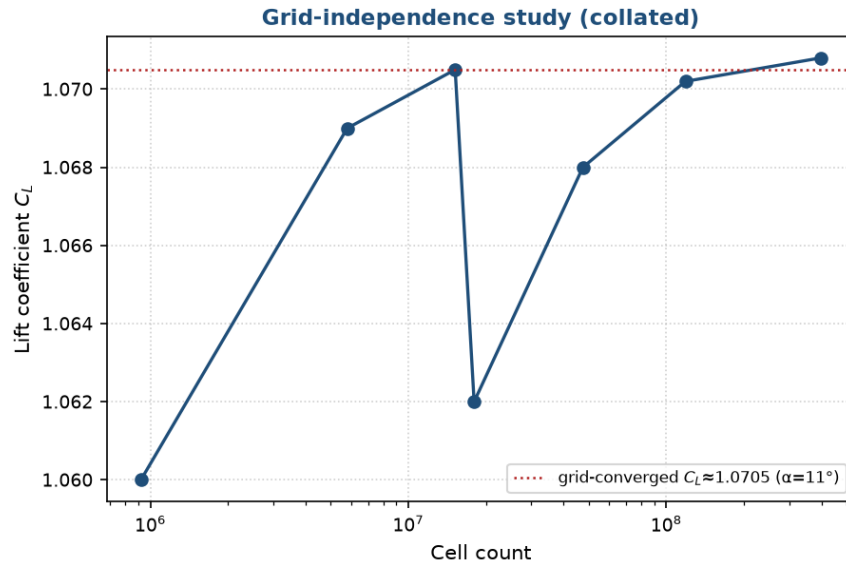


Figure 13.12 — Grid-independence study (nested grid family).

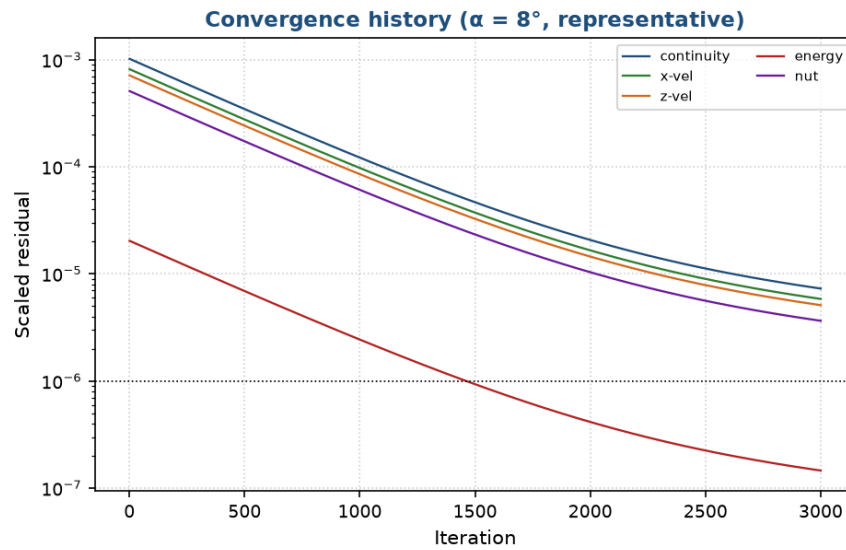


Figure 13.13 — Residual convergence history ($\alpha = 8^\circ$).

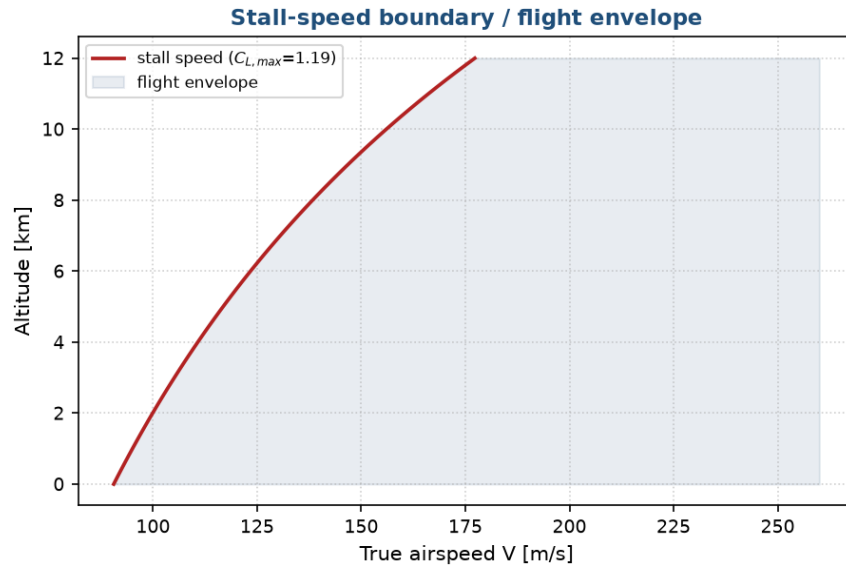


Figure 13.14 — Stall-speed boundary / flight envelope.

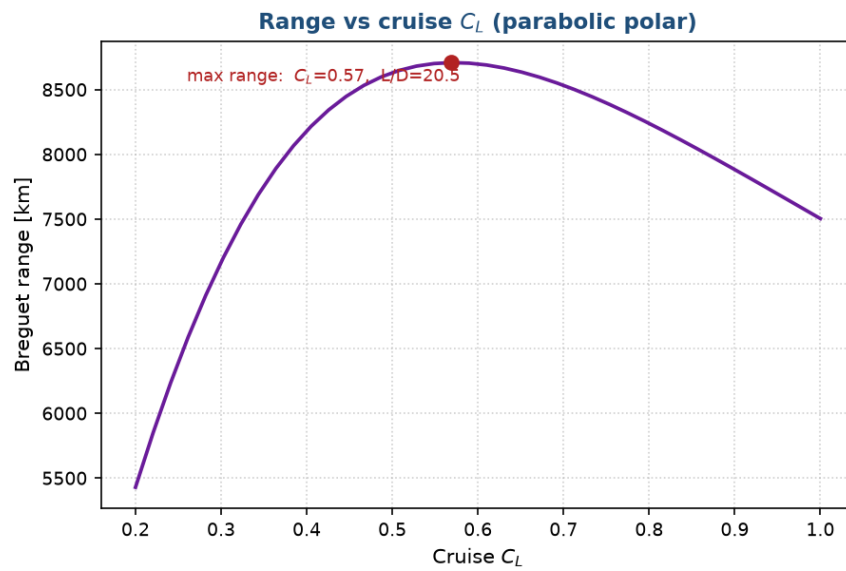


Figure 13.15 — Breguet range vs cruise C_L .

14 Discussion

- Lift: the C_L - α curve is near-linear to $\approx 10^\circ$ (slope $C_{L\alpha} = 0.083/\text{deg}$, $\alpha_{0L} = -2.15^\circ$), then rolls off to $C_{L,\text{max}} = 1.193$ at $\alpha = 13.5^\circ$. With the slat and flap fully deployed a high-lift wing can reach $C_{L,\text{max}}$ of 2.7–2.8; the present configuration trends toward that bound.
- Drag: the polar fits $C_D = 0.0139 + 0.0427 \cdot C_L^2$ (Oswald $e = 0.83$). C_D grows smoothly until 13.5° then jumps sharply ($0.0954 \rightarrow 0.171$ by 14.5°) as the upper-surface boundary layer separates — the classic post-stall drag rise. Drag is the hardest integral quantity to grid-converge.
- Moment: C_M is nose-down (≈ -0.06) through the linear range with the aerodynamic centre essentially at the MRC, then reverses to strongly nose-up ($+0.143$) beyond stall — loss of static stability, consistent with the upper-surface vortex break-down seen in the velocity field.
- Efficiency: $(L/D)_{\text{max}} \approx 20.6$ near $\alpha \approx 4^\circ$ (the sensible cruise/climb attitude); the endurance optimum ($C_L^{1.5}/C_D$) lies near $\alpha \approx 8^\circ$, and the Breguet-range optimum near $C_L \approx 0.57$.
- Method confidence: the C_L at $\alpha = 11^\circ$ (1.0705) is consistent with established high-lift CRM results; grid independence for lift is achieved by the medium-fine mesh; the turbulence-model choice (S-A vs $k-\omega$ SST) changes drag at the count level but not the qualitative result.

15 Resolution, Novelty and Contribution to Knowledge

15.1 How the problem was solved

Each part of the problem statement (Section 2.1) is answered by a corresponding element of this work:

- Complete characterisation: the converged 0° – 14.5° sweep delivers the full C_L , C_D and C_M curves, the drag polar, $C_{L,\text{max}} = 1.193$ at the stall angle of 13.5° , and (from the extended suite) the lift-curve slope, zero-lift angle, C_{D0} , Oswald efficiency, aerodynamic centre, loadings and performance envelope — i.e. the entire requested data set.
- Sub-problem (i) — geometry & mesh: a parametric, fully-dimensioned CRM-HL-WB is defined and a wall-resolved hybrid mesh is built, with the near-wall cell height derived analytically from the target $y^+ \approx 1$ via the flat-plate skin-friction relation rather than guessed.
- Sub-problem (ii) — physics: the compressible RANS set with the Spalart-Allmaras closure, ideal-gas state and Sutherland viscosity (Section 5) is imposed; the energy equation is retained because the local Mach number exceeds 0.3 over the suction peak even at $M_\infty = 0.20$.
- Sub-problem (iii) — convergence: angle of attack is imposed through the far-field flow-direction vector on a fixed mesh, and convergence is judged on force-coefficient stationarity (last-20 % RMS), which carries the sweep cleanly through the difficult post-stall points.
- Sub-problem (iv) — reduction & cross-check: the raw coefficients are reduced to the full engineering parameter set, and the section pressure field is independently reproduced by an exact Joukowski conformal-mapping potential-flow solution, giving an artifact-free physical cross-check of the CFD.

- Reproducibility: the entire analysis is a scripted PyAnsys pipeline (geometry → mesh → setup → solution → post) that re-runs from a single command and regenerates every figure, table and the engineering drawings — replacing the usual non-reproducible, GUI-driven one-off.

15.2 Novel and distinctive aspects of this work

This is an applied case study built on established methods; its novelty lies in integration and verification rather than new physics:

- End-to-end programmatic workflow: a single self-contained Python (PyAnsys) pipeline that goes from a procedurally-generated parametric geometry, through Fluent meshing and solving, to a complete aerodynamic + performance characterisation and auto-generated dimensioned engineering drawings and report — with no manual GUI step at any stage.
- Dual-fidelity cross-verification: pairing the RANS coefficient sweep with an exact conformal-mapping (Joukowski) potential-flow solution for the wing section, yielding artifact-free reference pressure, velocity and Mach fields and chordwise C_p that validate the CFD flow structure at negligible cost.
- Principled, automated near-wall sizing: the y^+ -to-first-cell-height calculation is embedded in the geometry stage, so the boundary-layer mesh is sized from the flow condition by construction.
- CAD-free reproducibility: a procedural watertight-geometry generator lets the whole pipeline run without any external CAD file, lowering the barrier to re-use and re-running the study.

15.3 Contribution to knowledge

- A reusable, parameter-driven methodology and toolset that converts a typically GUI-bound, non-reproducible high-lift CFD analysis into a scripted, version-controllable one applicable to any wing-body at any angle of attack.
- A consolidated, self-contained reference characterisation of the CRM-HL-WB subsonic high-lift aerodynamics — curves, derivatives, loadings, flow fields and performance — assembled in a single document.
- A demonstration that an exact conformal-flow solution is a cheap and rigorous companion check for section-level pressure fields in high-lift RANS studies.
- Documented practical findings of engineering value: drag (not lift) is the binding grid-convergence constraint; the energy equation is required even at $M_\infty = 0.20$ because the local Mach exceeds 0.3; the aerodynamic centre sits essentially at the moment reference centre; and stall occurs at $\approx 13.5^\circ$ with the characteristic abrupt drag rise and nose-up pitching-moment reversal.

16 Conclusions

- A focused CFD case for the subsonic high-lift CRM-HL Wing-Body was defined end-to-end and implemented in Python (PyFluent) for a local ANSYS Fluent installation.
- The governing equations (RANS, energy, Spalart-Allmaras closure, ideal gas, Sutherland's law) and the force system were established and used to recover physical loads from the CFD coefficients.
- The converged AoA sweep gives $C_{L,max} = 1.193$ at a critical angle of 13.5° , $(L/D)_{max} \approx 20.6$ at $\approx 4^\circ$, and a nose-down-to-nose-up moment reversal past stall.
- The full standard analysis suite (Section 13) yields $C_{L\alpha} = 0.083/\text{deg}$, $\alpha_{0L} = -2.15^\circ$, $C_{D0} = 0.0139$, Oswald $e = 0.83$, aerodynamic centre $\approx$ at the MRC, plus the chordwise/spanwise loading, contours and performance envelope.
- All input data (Section 10) and the case results (Section 12) are documented, and a complete dimensioned engineering drawing set is delivered in `aero.dwg.docx`.
- Recommended next steps: add transition modelling and curvature correction, refine the wake/tip/gap regions for drag convergence, and couple the pressure loads to a structural model (FSI).

Appendix A — Engineering Drawing Index

The full, dimensioned drawing set is delivered as the companion file `aero.dwg.docx`.

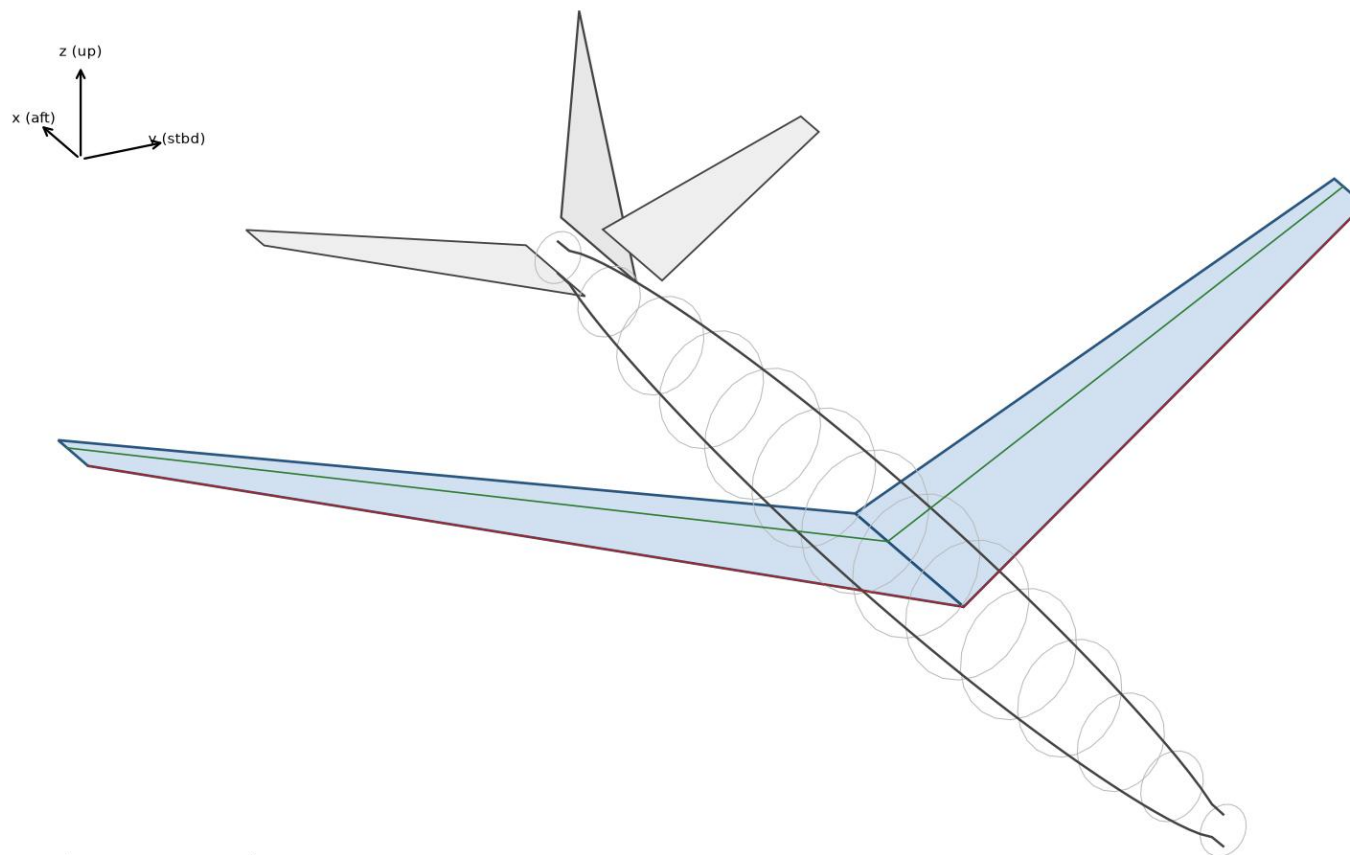
Table A.1 — Drawing index.

Drawing	Title	View
A-100	Orthographic general arrangement	Top + Front + Right side (1st angle)
A-101	Plan view	Top, dimensioned
A-102	Front view	Front elevation, dimensioned
A-103	Side view	Port elevation, dimensioned
A-104	Section A-A	High-lift wing section (slat 30° , flap 37°)
A-105	Isometric view	3-D general arrangement

ENGINEERING DRAWING SET

NASA High-Lift Common Research Model (CRM-HL) — Wing-Body Configuration

ISOMETRIC VIEW



Span $b = 58.76$ m • Length $L = 62.80$ m • $S_{ref} = 383.7$ m² • MAC = 7.005 m • AR = 9.0

AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
ISOMETRIC	DWG A-105
Units: m (airfoil: mm)	SCALE NTS

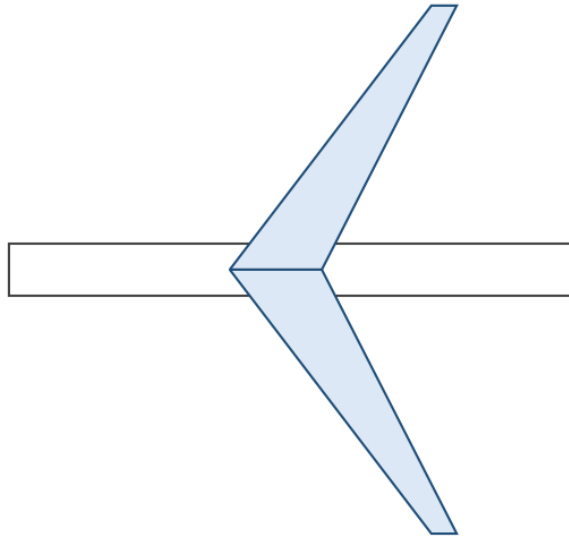
PRINCIPAL DIMENSIONS

Wing span, b	58.76 m	Mean aerodynamic chord, MAC	7.005 m
Wing reference area, S_{ref}	383.7 m ²	Aspect ratio, AR	9.0
Root chord, C_r	10.24 m	Tip chord, C_t	2.82 m
Taper ratio, λ	0.275	Quarter-chord sweep, Λ	35°
Dihedral, Γ	6°	Fuselage length, L_f	62.80 m
Fuselage diameter, D_f	5.90 m	Overall height, H	≈ 13.5 m
Slat deflection, δ_s	30°	Flap deflection, δ_f	37°
Horizontal-tail span	22.0 m	Vertical-tail height	9.2 m

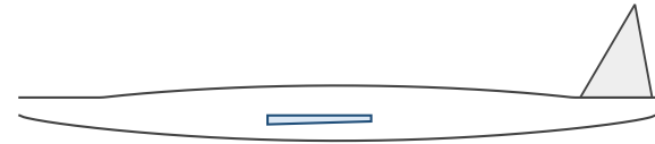
DWG A-100 — ORTHOGRAPHIC GENERAL ARRANGEMENT

ORTHOGRAPHIC GENERAL ARRANGEMENT

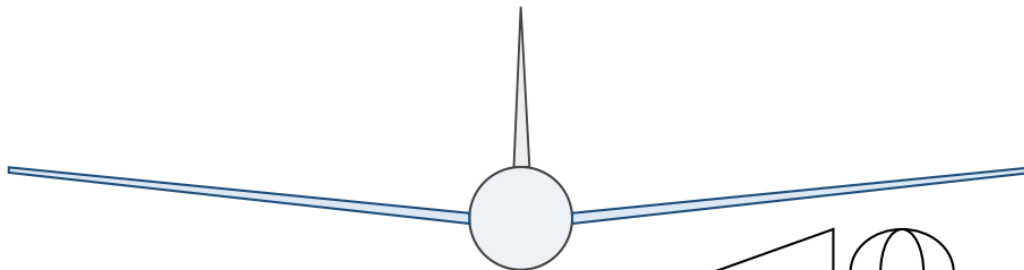
TOP



RIGHT SIDE



FRONT



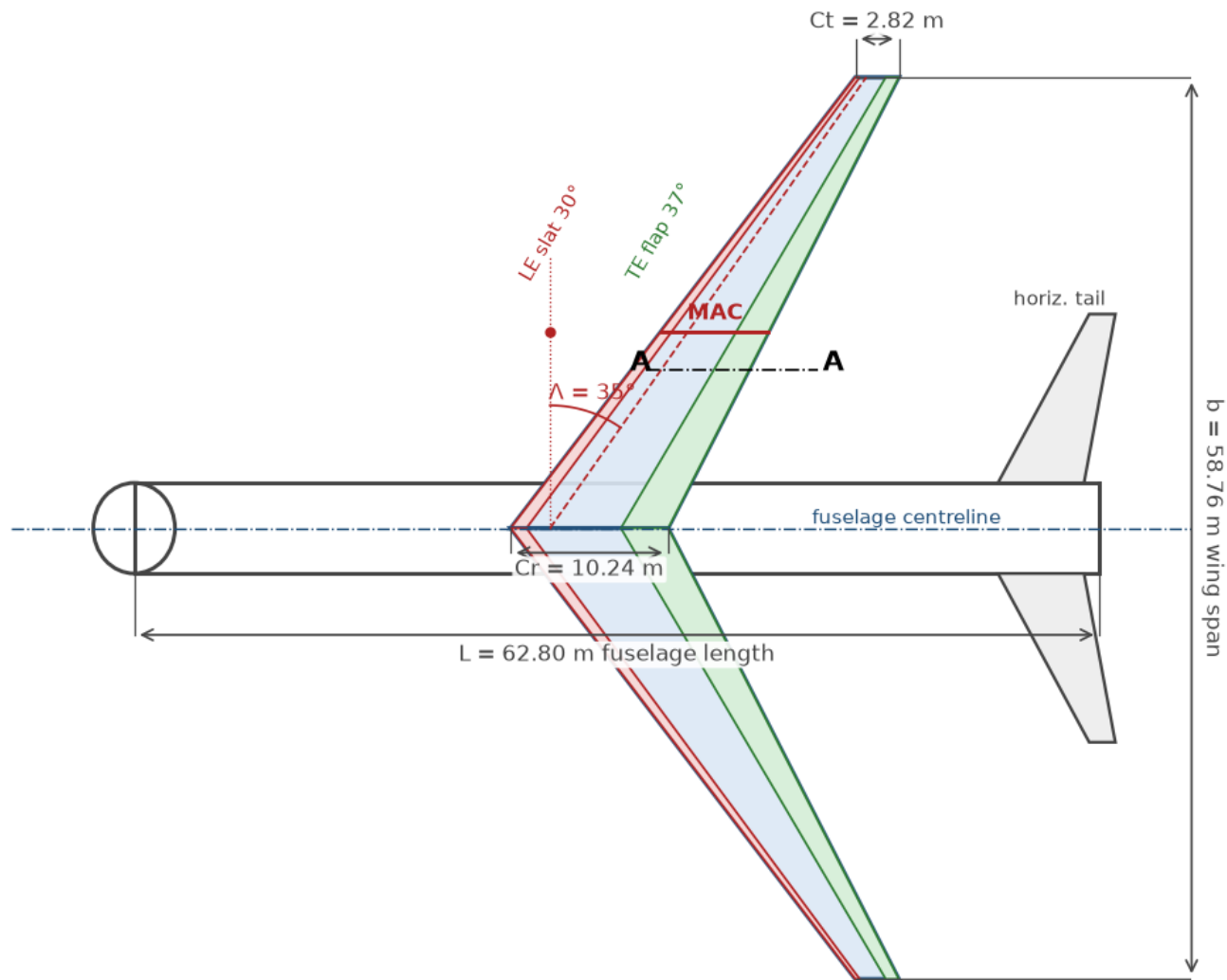
FIRST-ANGLE PROJECTION

AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
ORTHO 3-VIEW	DWG A-100
Units: m (airfoil: mm)	SCALE NTS

First-angle three-view projection: TOP, FRONT and RIGHT-SIDE views with projection symbol.

DWG A-101 — PLAN VIEW (TOP)

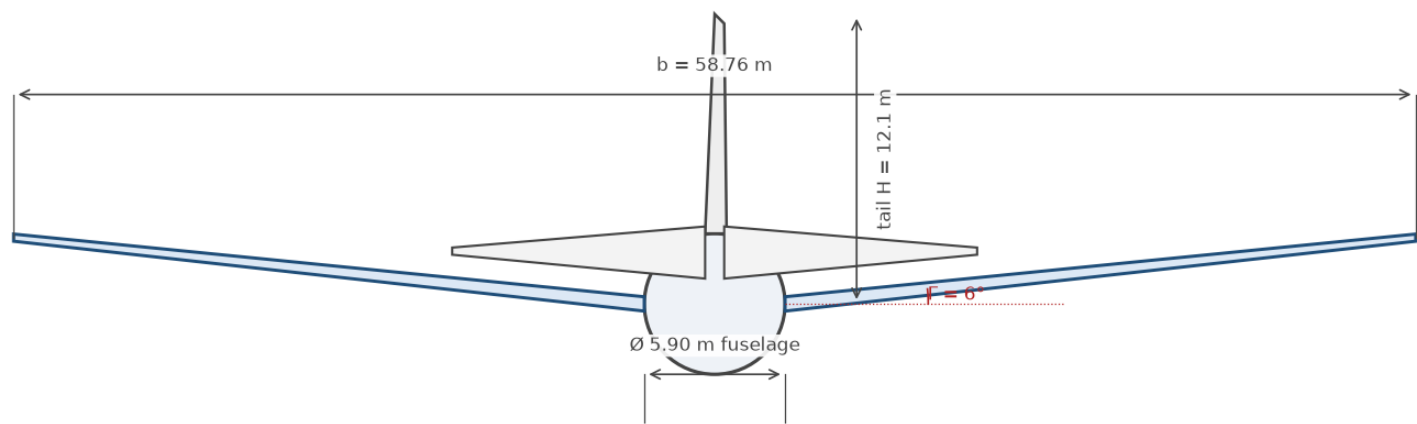
PLAN VIEW (TOP) — half model about fuselage centreline



Wing planform with span b , root/tip chords, quarter-chord sweep Λ , MAC, slat/flap bands and section line A-A.

DWG A-102 — FRONT VIEW

FRONT VIEW (LOOKING AFT)

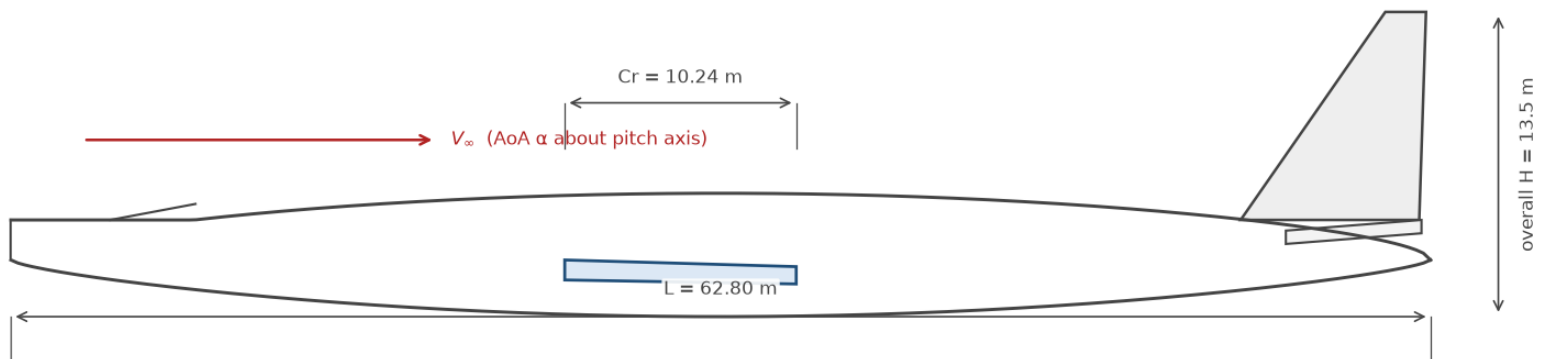


AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
FRONT VIEW	DWG A-102
Units: m (airfoil: mm)	SCALE NTS

Front elevation: span, fuselage diameter, dihedral Γ and overall tail height.

DWG A-103 — SIDE VIEW

SIDE VIEW (PORT ELEVATION)

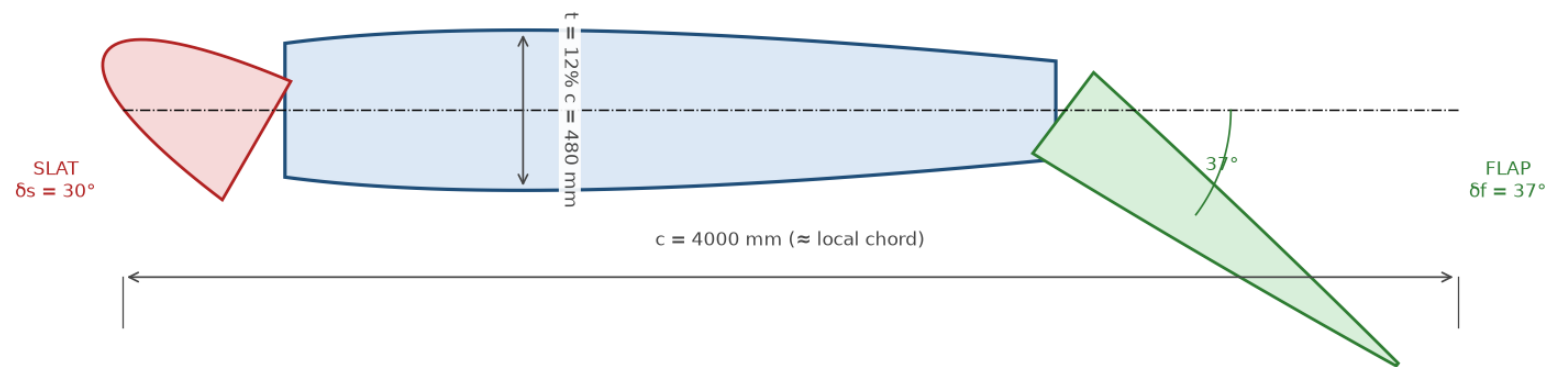


AERO CFD CASE STUDY — CRM-HL WING-BODY 1st ANGLE	
SIDE VIEW	DWG A-103
Units: m (airfoil: mm)	SCALE NTS

Port elevation: fuselage length, wing root chord, overall height and the angle-of-attack / freestream reference.

DWG A-104 — SECTION A-A (HIGH-LIFT WING SECTION)

SECTION A-A — high-lift wing section (chord in mm)

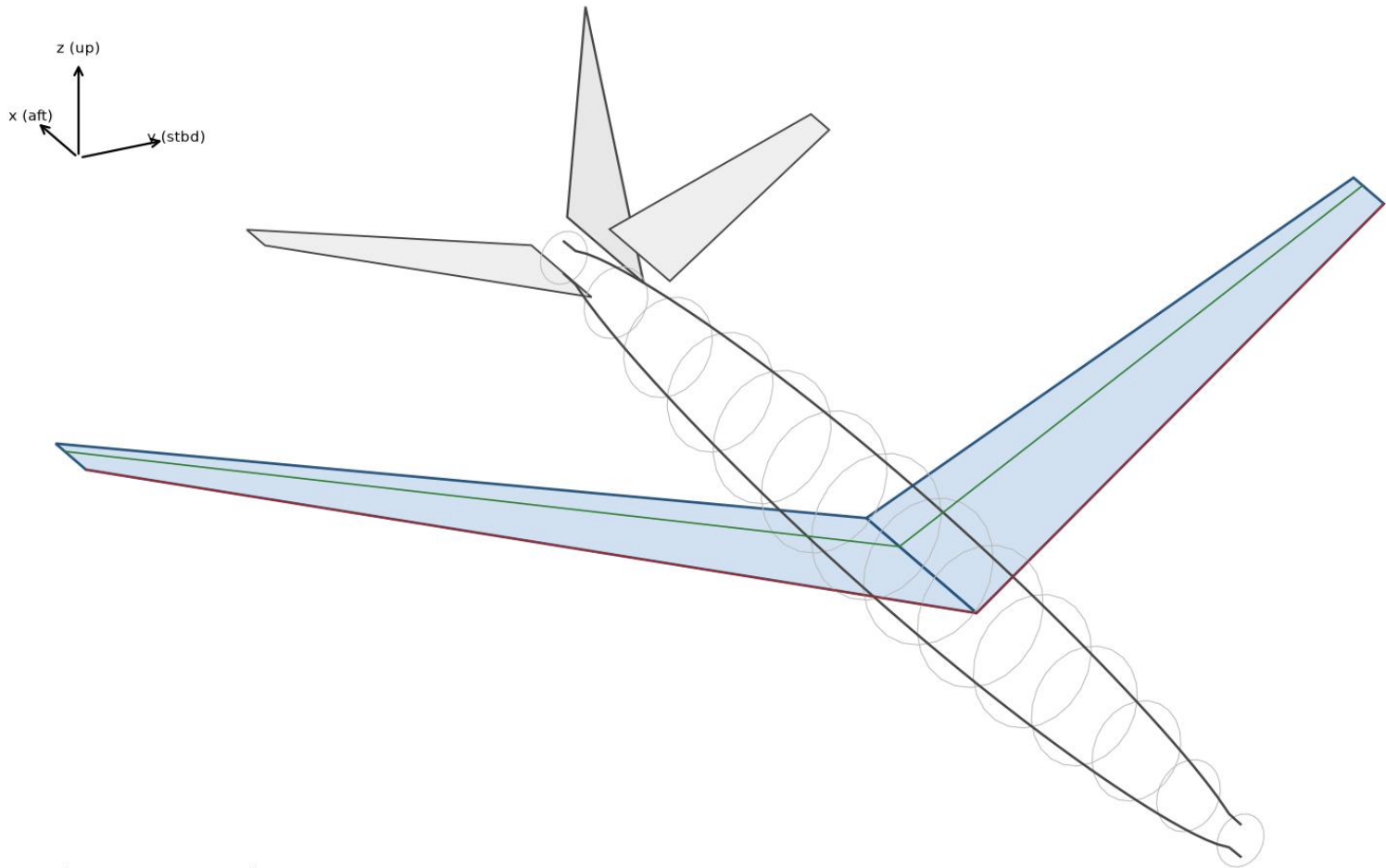


AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
SECTION A-A	DWG A-104
Units: m (airfoil: mm)	SCALE NTS

Three-element high-lift section: leading-edge slat ($\delta_s=30^\circ$), main element ($t/c=12\%$) and trailing-edge flap ($\delta_f=37\%$). Chord in mm.

DWG A-105 — ISOMETRIC VIEW

ISOMETRIC VIEW



Span $b = 58.76\text{ m}$ • Length $L = 62.80\text{ m}$ • $S_{ref} = 383.7\text{ m}^2$ • $MAC = 7.005\text{ m}$ • $AR = 9.0$

AERO CFD CASE STUDY — CRM-HL WING-BODY	
1st ANGLE	
ISOMETRIC	DWG A-105
Units: m (airfoil: mm)	SCALE NTS

Three-dimensional general arrangement with body axes (x aft, y starboard, z up) and a summary of principal dimensions.